The Mandarin Model on the Ground: Career Incentives and Local Development in Chinese Cities*

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July 6, 2026

Abstract

The mandarin growth model emphasizes how politicians' career incentives have shaped China's economic growth (Song and Xiong, 2026). We identify how these incentives affect local development in Chinese cities. Our empirical design exploits appointments of provincial superiors who share city leaders' birthplaces as exogenous relaxations of career incentives, which sever the link between growth and promotion. We document that these city leaders reallocate land toward residential use and away from industrial and commercial uses, lower residential land prices, and curb local-government borrowing, with no detectable change in total land supply. The results reveal how career incentives redirect local development.

*We are deeply grateful to our advisors Arvind Krishnamurthy, Hanno Lustig, Monika Piazzesi, and Martin Schneider, Chris Tonetti, and Michael Boskin for their patient guidance and generous support. We thank Adrien Auclert, Rebecca Diamond, Liran Einav, Zhiguo He, Thomas Helgerman, Ali Yurukoglu, and seminar participants at AREUEA National Conference (2020), Stanford Economics, and Stanford GSB for helpful comments and feedback. We thank Chuanzhang Li and Xiao Zhang (Chinese Academy of Social Sciences) and Wei Xiong (Princeton University) for sharing data. An earlier version of this paper was titled "The Political Economy of China's Housing Boom". All errors are our own.

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1 Introduction

The *mandarin model* highlights political incentives behind China's economic growth: Chinese politicians have broad economic discretion and are rewarded for local economic performance (Song and Xiong, 2026). At the city level, it implies that the city leaders' career incentives shape local economic decisions, including how land is allocated across competing uses and how much debt local governments accumulate. Causally identifying this mechanism is difficult because career incentives are not directly observed and are themselves endogenous to the outcomes they may affect.

We provide causal evidence that political career incentives shape local economic development in Chinese cities. Our empirical design exploits variation in city leaders' career incentives. In China's promotion system, most prefectural city leaders are evaluated by provincial superiors, with economic performance playing a central role in these top-down evaluations (Li and Zhou, 2005). The provincial superiors are appointed from above, at times and for reasons plausibly unrelated to any given city's land and housing markets. Building on evidence that social and political ties shape political selection in China (Fisman, Shi, Wang and Wu, 2020, Chu, Fisman, Tan and Wang, 2021), we use hometown ties, that is, when newly appointed superiors and incumbent city leaders share the same birthplace, as a source of variation in career incentives.

GDP growth significantly predicts promotion for city leaders, unless they share a hometown tie with their provincial superior. For untied leaders, a one-standard-deviation rise in annualized growth increases promotion probability by about 13 percentage points, sizable against a 35% promotion rate. A hometown tie, by contrast, renders economic performance irrelevant. Conditional on GDP growth, demographics, and other promotion-relevant covariates, a hometown tie independently raises the odds of advancement.

Relaxing career incentives affects city land supply and prices, as well as local government borrowing. City leaders who experience hometown ties shift the composition of land supply toward residential use and away from both industrial and commercial land, raising the residential share by six to seven percentage points, and they slow local government debt accumulation, leaving the outstanding stock about 31% below its no-tie counterfactual in the year after the hometown tie forms. Because local debt was compounding at roughly 36% a year over this period, this amounts to about one year of forgone new borrowing. At the same time, we do not detect a change in the *total* quantity of land supplied, suggesting the tie works through the compositional changes of land uses. The changes in the supply of land show up in prices. Residential land prices fall by about 19%, and industrial land prices rise by about 15% as the hometown tie forms. We also document supporting evidence on real estate prices. House-price growth slows by roughly 5% per year, against an average of about 13% a year over the sample.

The reallocation also propagates into the real economy. We show that in the two years after a tie forms, the city's industrial activity contracts. Foreign direct investment falls, as do industrial employment and the urban population, while residential housing expands and aggregate output actually increases. Further decomposition shows the post-tie growth is driven by construction and the material demand it spurs.

We interpret these findings as the city-level footprint of Song and Xiong's "Mandarin" account of Chinese growth (Song and Xiong, 2026), in which the political promotion system shapes the character of local development through decisions on land supply and debt accumulation. Career incentives steer local growth toward the industrial activity that the promotion tournament credits to a leader, such as foreign investment, factories, and debt-financed investment.

Related literature. The paper makes three main contributions. First, it provides causal micro-evidence for the career-incentive mechanism that the macro literature on China's mandarin model places at the center of how local economies are governed (Song and Xiong, 2026). That literature works largely through theory and aggregate accounting. We supply the corresponding evidence at the individual-leader level, isolating an exogenous, within-leader relaxation of career incentives and tracing its effect on land supply, land and housing prices, local government debt, and the local real economy. In doing so, we connect to a broad literature on political connections and economic outcomes, in cross-country data (Faccio, 2006), in developing countries (Khwaja and Mian, 2005, Claessens, Feijen and Laeven, 2008, Acemoglu, Hassan and Tahoun, 2018), and in developed economies (Goldman, Rocholl and So, 2009, Cooper, Gulen and Ovtchinnikov, 2010, Amore and Bennedsen, 2013, Akey, 2015, Acemoglu, Johnson, Kermani, Kwak and Mitton, 2016, Schoenherr, 2019). In China specifically, personal ties shape outcomes ranging from promotion (Jia, Kudamatsu and Seim, 2015) and trade credit (Kong, Pan, Tian and Zhang, 2020) to favorable treatment in elite scientific selection (Fisman, Shi, Wang and Xu, 2018), lenient government monitoring (Chu, Fisman, Tan and Wang, 2021), corporate tax breaks (Cao, Cheng, Xia, Xie and Zeng, 2024), and protection from anticorruption enforcement (Griffin, Liu and Shu, 2022). At the top of the political system, where connections invite scrutiny, ties can instead carry a penalty in selection to the Politburo (Fisman, Shi, Wang and Wu, 2020), a contrast compatible with favoritism operating at the local level we study. Most evidence on connections and promotion concerns provincial leaders and above, whereas we focus on city leaders at the prefecture tier where land is allocated and debt is incurred. Similar to our setting, Wang, Zhang and Zhou (2020) model city leaders as "city developers" and show that stronger career incentives push them to expand cities outward at low density to raise land revenue, while Henderson, Su, Zhang and Zheng (2022) structurally estimate the weight leaders place on growth relative to resident welfare that drives a wedge between industrial and residential land supply. These studies establish that career incentives shape the quantity, spatial form, and modeled preferences behind land development. Our outcome is the composition of land use across its industrial, commercial, and residential parcels, not its quantity or density. Our variation is an exogenous, within-leader relaxation of the growth-promotion incentive, the hometown tie, rather than cross-sectional career-incentive proxies or a calibrated preference. Moreover, we follow the resulting shift through prices, debt, and the recomposition of the local economy, which are city-level evidence of the mandarin growth model.

Second, this paper contributes to the literature on the demand and supply factors behind

China's housing boom. Previous studies have examined demand factors extensively, such as speculative investment, status competition, credit expansion, and urbanization, as well as fundamental factors such as population, wages, income, and construction costs (Wang and Zhang, 2014, Chow and Niu, 2015, Fang, Gu, Xiong and Zhou, 2016, Wu, Gyourko and Deng, 2016, Chen and Wen, 2017, Wei, Zhang and Liu, 2017, Chen and Zhang, 2023). A smaller literature emphasizes the supply side, noting that local governments control urban land supply and that supply constraints may be central to China's housing boom (Glaeser, Huang, Ma and Shleifer, 2017, Wu, Gyourko and Deng, 2016). Our contribution is to identify a political-career determinant of this supply side. Prior work shows that China's political institutions matter for land and housing markets (Cai, Henderson and Zhang, 2013) and Chen and Kung (2019). Most closely related, Chang, Wang and Xiong (2026) show that local governments in China actively manage residential land and house prices. We study a complementary mechanism, how city leaders' career incentives, and their relaxation through hometown ties, shape land-use composition, local-government borrowing, and house-price growth.

A separate literature emphasizes local governments' *fiscal* incentives as a determinant of land supply, pointing to fiscal pressure and land-based financing (Gong, 2012, Li, Hong and Huang, 2013, Han and Kung, 2015, Liu and Xiong, 2020). He, Nelson, Su, Zhang and Zhang (2025) show that the large residential-industrial price gap is substantially rationalized by the future tax revenues that industrial land generates. We view our channel as complementary to these accounts. In addition to the fiscal fundamentals they emphasize, we isolate the distinct, political-career determinant of land allocation causally, using the hometown tie as exogenous within-leader variation a correlational design cannot deliver. The career-incentive channel also builds on the literature on China's political tournament. Politicians are ranked by their relative performance among their peers, the principle of yardstick competition (Shleifer, 1985, Maskin, Qian and Xu, 2000), for which Chen, Li and Zhou (2005) provide direct evidence from provincial-leader turnover. We show that evaluating city leaders on relative economic performance within provinces has important implications for land supplies and house-price growth.

Third, our findings speak to the rapidly growing literature on local government debt and land finance in China. A large body of work documents the scale and risks of local government borrowing, much of it intermediated by local government financing vehicles and collateralized by land (Ambrose, Deng, Li and Wu, 2024, Song and Xiong, 2018, Chen, He and Liu, 2020), and links land-based revenue to local fiscal and political behavior (Chen and Kung, 2016, Li, Hong and Huang, 2013). A parallel strand debates whether local spending and debt crowd private investment in or out (Su, 2025, Huang, Pagano and Panizza, 2020). The literature has emphasized fiscal-pressure and stimulus-driven accounts of the debt buildup (Bai, Hsieh and Song, 2016), and recent regional evidence shows that shortfalls against growth targets trigger additional land sales and borrowing (Chang, Wang and Xiong, 2025), while Song and Xiong (2026) argue that career incentives foster over-leverage at the aggregate level. We provide direct, city-level causal evidence for the latter. Because debt-financed investment is itself an instrument for delivering the GDP growth on which promotion depends, a relaxation of career incentives through a hometown tie leads leaders to curtail borrowing. Political career incentives are an underappreciated driver of local government debt accumulation, complementary to the fiscal-capacity and stimulus channels emphasized in prior work.

2 Institutional Background

In this section, we describe two institutional features that shape how a city leader allocates land: China's cadre evaluation system and the local government's monopoly over urban land.

2.1 Career Incentives in the Party Hierarchy

The Chinese Communist Party governs through a four-tier hierarchy running from the central Politburo down through provincial and prefecture committees to the counties (the political pyramid of Figure 11). We focus on the provincial and prefecture tiers. A prefecture's party secretary, hereafter the city leader, heads the city government and sets local policy, including land allocation. Party personnel are by law elected by congresses, but in practice the committee one level up controls appointment, promotion, and demotion. A city leader's career rests with the provincial leadership (Li and Zhou, 2005).

The evaluation system rewards economic performance. Provincial superiors evaluate city leaders by GDP growth, a yardstick competition that has made local output the test of political advancement (Chen, Li and Zhou, 2005, Song and Xiong, 2026). Evaluation is not purely mechanical, however. Personal ties enter alongside performance, and a connection to a higher-ranking official can outweigh a stronger professional record. Among such ties the shared-hometown bond holds a special place. The term *lǎoxiāng*, people from the same place whether or not they have ever met, denotes a recognized claim to mutual obligation and favor (Fisman, Shi, Wang and Xu, 2018). When a newly appointed provincial superior is a city leader's *lǎoxiāng*, advancement ceases to hinge on out-growing one's peers, which we document and exploit.

2.2 The Urban Land Market and the City Leader's Choice

All urban land in China belongs to the state, and since the 1998 Land Management Law local governments hold the right to lease it. A city government is therefore the monopoly supplier of land in its jurisdiction, selling multi-decade usufruct rights to firms and developers (Liu and Xiong, 2020). How much rural land a city may convert to urban use is not its own decision. The Ministry of Land and Resources sets a national quota, fixed in a multi-year land-use master plan and implemented through annual plans, and allocates it down through the provinces to the cities (Wang, Zhang and Zhou, 2020, Liu and Xiong, 2020). That quota is in turn disciplined by a farmland-protection target. A binding national floor of roughly 1.8 billion mu (1 mu $\approx$ 0.165 acres) of arable land, the farmland red line, requires each city to offset any conversion of farmland to urban use with comparable farmland created inside its own jurisdiction, which raises the cost of expanding the urban territory and makes the supply of developable land a constrained choice (Yu, 2024). Because that total is set on a multi-year horizon, the city leader's effective choice variable within a typical four-year term window is how that total is divided among industrial, residential, and commercial use, with residential and commercial parcels, as well as industrial parcels post 2007, sold by public auction to address corruption in land markets (Cai, Henderson and Zhang, 2013). Within the city, land use is planned by a land-reserve and allocation committee headed by the city leader (Wang, Zhang and Zhou, 2020). Figure 13 shows that industrial and residential uses absorb the bulk of supply over our sample period.

This composition decision embeds a sharp tension. Industrial land is kept cheap. Cities routinely grant it at or below cost to attract manufacturers, who bring the investment, employment, and industrial output. Residential land, far more valuable, is the principal source of land-sale revenue. Over 2004–2015 residential land prices rose several-fold while industrial prices stayed nearly flat (Liu and Xiong, 2020). Tilting supply toward industry therefore trades current land value for growth credentials.

Land revenue, and the debt built upon it, is the second reason the allocation is politically charged. The 1994 tax-sharing reform left local governments with heavy spending obligations but a smaller share of tax revenue. It pushed them toward land sales, whose revenue grew rapidly over the period (Figure 12), and, after 2008, toward land-collateralized borrowing through local financing vehicles (Han and Kung, 2015, Liu and Xiong, 2020, Song and Xiong, 2018). Land-related revenue has grown to roughly a third to half of local budgets (Figure 14). This fiscal-incentive account of land allocation is developed recently by He, Nelson, Su, Zhang and Zhang (2025). We contribute with new evidence that the career incentives of city leaders play a key role in understanding land revenue and debt across Chinese cities.

2.3 Relaxing Career Incentives and Implications

Consider a city leader who allocates the city's land across industrial, residential, and commercial uses and decides how much debt-financed investment to undertake. Residential land yields large upfront sale revenue, while industrial land yields modest upfront revenue but a stream of future tax payments from the firms it hosts, and borrowing finances the infrastructure that supports development (He, Nelson, Su, Zhang and Zhang, 2025). At the same time, promotion in the cadre system rewards the measured economic performance of one's jurisdiction (Li and Zhou, 2005, Song and Xiong, 2026). Industrial production and debt-financed infrastructure register directly in GDP and investment, and are visibly attributable to the leader's own effort, in a way that the real-estate construction a residential build-out generates is not, a premise the promotion evidence in Section 4 bears out. A leader who weights promotion heavily therefore has reason to tilt land toward industrial and commercial uses and to borrow aggressively, beyond fiscal concerns alone. When promotion is scored on the achievements a superior can measure and credit, the leader tilts toward the creditable industrial and commercial margins. The premise has direct counterparts in the literature. Governments over-provide public goods whose outcomes evaluators can observe and attribute to official competence (Mani and Mukand, 2007). Within China's cadre system specifically, city spending tilts toward visible projects such as roads, bridges, and landscaping, at the expense of valuable but harder-to-observe ones, for promotion-eligible leaders and around evaluation windows (Wu and Zhou, 2018). And the promotion mechanism demonstrably rewards short-horizon, readily evaluable investments over education and health (Persson and Zhuravskaya, 2016).

Our research design isolates the career incentive. When a city leader's superior is replaced by a newly appointed official who shares the leader's birthplace, a hometown tie forms, a salient and well-documented basis for favoritism among Chinese officials (Fisman, Shi, Wang and Xu, 2018). We show in Section 4 that such a tie raises the city leader's promotion probability while breaking the link between promotion and GDP. Because the superior's appointment is decided at the central level, at a time and for reasons unrelated to any individual city's land trajectory, a tie acts as a plausibly exogenous relaxation of the GDP-promotion incentive for a city leader. A leader freed from growth pressure should rebalance land toward residential use. The reallocation should affect prices, with residential land becoming cheaper and industrial land more expensive. Furthermore, because the same growth motive drives debt-financed investment, tied leaders should borrow less. The reallocation should also affect the real economy itself through land allocation. Industrial growth should wind down, while residential construction activity should expand.

3 Data

In this section, we first detail the data collection process of Chinese politicians' biographical information and career record, followed by procedures to construct the measure of political promotion. We then explain city-level real estate, land supply, and other macroeconomic data from various sources and present relevant descriptive statistics.

3.1 Political Data

We combine two sources of data to construct our sample on Chinese politicians. Our main source of political data, the Chinese Political Elite Database (CPED), contains extensive demographic information for Politburo/provincial/prefectural politicians.

Politicians in CPED include all city party secretaries and mayors, provincial party secretaries and governors, and all other full and alternate CCP Central Committee members between 1995 and 2015.[1]

CPED records the start and end dates, workplaces, and political ranks of every job assignment in each politician's career, hand-collected from government websites, yearbooks, and other credible sources. We verify it against the Provincial and City Leader Database maintained by CSMAR and resolve any disparities manually against official government sources.

In short, we observe the universe of city and province party committee leaders who assumed office between 2001 and 2015, and for each we obtain biographical information and the entire career path. Our analysis focuses on the 636 leader terms completed within the 2003–2015 window. We begin in 2003 because the marketization of urban land supply took off that year (Liu, Cao, Yan and Wang, 2016), the reform that makes the allocation of land a meaningful margin of a city leader's choice. The promotion regressions are estimated on the 411 of these terms that have a measured over-term GDP growth rate and contribute identifying variation within their province-year cells, summarized in Table 1.

[1] We focus on the period before 2015 because since 2016, the Party Central Committee led by Xi Jinping has designated "housing is for living, not for speculation" as a long-term national strategy, and as a result, various local policy restrictions have been implemented to curb housing prices.

3.1.1 Promotion

The Chinese government has a top-down administrative structure. Instead of elections, promotions of provincial and prefectural politicians are decided by government officials of higher political ranks. The power structure is the pyramid described in Section 2: the 25 Politburo members at the top, the provincial CCP committees below (13 members each, 34 provincial divisions in total), and the prefecture-level committees at the base (10–11 members each, 293 official prefecture cities in total). Leaders are generally promoted up this ladder level by level, and the promotion tournament is local. A city leader's performance is evaluated relative to other city leaders from the same province (Chen, Li and Zhou, 2005, Yao and Zhang, 2015). We focus on city leaders in this paper and, following Wang, Zhang and Zhou (2020), measure promotion by movement in political ranks. Based on the observed career paths of politicians in the data, we define a government official as promoted if he is appointed to a new position with a higher political rank at the end of term in office.

Fifty-six city leaders in our sample were dismissed during their political careers, commonly for corruption, which could introduce biases to our estimates (Cai, Henderson and Zhang, 2013, Fang, Wu, Zhang and Zhou, 2022). We exclude these politicians from the main analysis.

3.1.2 Promotion for Local Communist Leaders

There has been an extensive literature on the Chinese government's personnel control. Li and Zhou (2005) find that for the turnover of provincial leaders, GDP performance, age, central connections, and years in office are among the most influential factors. At the city level, Yao and Zhang (2015) show that individual leaders matter for growth and that abler leaders are more likely to be promoted. Our promotion regressions condition on a similar set of characteristics.

3.2 Real Estate Data

We obtain urban planning and land sales data aggregated at city level from China Real Estate Index System (CREIS) database maintained by China Index Academy. CREIS collected land transaction data from the Ministry of Land and Resources, and local Land Reserve Centers.

Information available in CREIS includes acreage, price, and land-use type for each parcel of land. These China Index Academy land-transaction records are widely used in research on China's land market (Wang, Zhang and Zhou, 2020, Chang, Wang and Xiong, 2026). We measure land supply by the shares of land the city government lists for sale, the supplied shares, and land prices by the realized auction-clearing prices on the parcels that sell. Supply is the leader's choice variable, and the transaction price is from the auction that follows. We compute these shares by building area, the floor space a parcel permits, and not by land area, because floor space better reflects the economic activity a use generates, especially for residential land, which is built at far higher density than industrial land. The land-share results are similar when measured by land area.

3.3 Macroeconomic Series

National Bureau of Statistics of China (NBS) maintains annual macroeconomic series at the city level. For each city, we collect population (measured by usual residence), government in-budget revenue and expenditure, total city-wide deposits, average wage, GDP, secondary- and tertiary-industry GDP, fixed investment, floor area sold for residential building, and residential housing prices by year during 2003–2015. Summary statistics for the land-market variables are reported in Table 2 and for the macro series in Table 3, and all price-related variables are normalized to a base year using the CPI published by NBS. The land-market regressions draw on prefecture cities with CREIS land-transaction coverage, up to 191 cities after excluding the four direct-controlled municipalities, of which 179 enter the baseline land panel once dismissed leaders and missing outcomes are excluded (Table 2, and pooling the municipalities back in is examined in Appendix Table 12). Because CREIS's systematic coverage of city land markets widens over the sample period, the land panel is unbalanced and concentrated in the better-covered years from the late 2000s on. The land-share and land-price regressions include roughly 500 city-years, and sample sizes vary across outcomes with data availability.

For house prices, our primary measure is the constant-quality residential price index of Fang, Gu, Xiong and Zhou (2016), who construct hedonic, repeat-sales-style indices from transaction-level data for roughly 120 prefecture cities. As Fang, Gu, Xiong and Zhou (2016) document, the official NBS average-price series conflate genuine price changes with shifts in the size, quality, and geographic mix of transacted units. A constant-quality index is needed to measure house-price growth accurately. We therefore use the Fang, Gu, Xiong and Zhou (2016) index wherever it is available (through 2012) and splice on the NBS city-level residential price series for the remaining years of our sample, for which official coverage and quality have improved (Liu and Xiong, 2020). House-price growth is the annual log change in this spliced index.

3.4 Local Government Debt

Our measure of local government debt is the city-level panel of total debt owed by local government financing vehicles constructed by Huang, Pagano and Panizza (2020). Municipalities in our sample period could not borrow from banks or issue bonds in their own name. Local borrowing ran through these vehicles, which the city capitalizes with transferred assets, usually land. The panel is assembled from the vehicles' balance sheets, which any vehicle applying to issue a bond must disclose for the current year and at least the three preceding years. It therefore captures total interest-bearing debt, which includes short-term and long-term borrowing, notes payable, bonds outstanding, and other current liabilities. Aggregated nationally, the series tracks the official National Audit Office figures to within five percent in 2012 and 2013 and matches their distribution across provinces. The outcome in our regressions is the logarithm of the outstanding debt stock in a city-year. The debt regressions also control for the average coupon on the city's bond issues that year (the Average Bond Coupon row in Table 10), a summary measure of the cost of credit the city faces. Debt is a stock with substantial inertia (the within-city-term first-order autocorrelation is 0.60), which motivates the lagged-tie timing convention discussed in Section 4.

3.5 Real-Economy Outcomes

Section 4.4 examines a set of real-economy outcomes drawn from CEIC's prefecture-level China statistical series, merged to our sample by province and city name, including foreign direct investment (utilized value, contracted value, and the number of contracts), sectoral employment and output, non-agricultural population, and residential floor space sold. The interpretation checks of Section 4.3.5 draw on two further sources. We measure net fiscal transfers from above with the ratio of general-budget expenditure to own revenue, both already in the analysis panel, and we measure municipal-infrastructure investment by funding source, with central appropriations itemized separately, from the Urban Construction Statistical Yearbook for 2006 through 2015. For the sectoral decomposition of GDP we draw city value added by industry from provincial statistical yearbooks compiled by EPS China Data.

4 Empirical Results

We begin by showing that a hometown tie relaxes a city leader's career incentive. We then investigate how the relaxed incentive reshapes the allocation of land, whether that response operates through composition or scale, and how it affects land and house prices and local borrowing. We then provide robustness checks, including a randomization-inference placebo, a specificity test against other social ties, and heterogeneity by promotion stakes, and two alternative interpretations. Finally, we show the effects on the real economy.

4.1 The Tie Decouples Promotion from Growth

A hometown tie alters the career incentive a city leader faces. The promotion of city leaders rewards economic performance. If a hometown tie blunts that link, the growth-promotion relationship would weaken or disappear when a tie is present . We therefore regress a leader's end-of-term promotion outcome on GDP performance, a hometown-tie indicator, and their interaction,

$$Y_{i,t} = \beta_0 + (\beta_1 + \beta_2 \textit{Hometown Tie}_{i,t}) \times \text{GDP Growth}_{i,t} + \beta_3 \textit{Hometown Tie}_{i,t} + \beta_4 X_{i,t} + \epsilon_{i,t} \quad (4.1)$$

where $Y_{i,t}$ is a promotion dummy for leader i whose term ends in year t, and each observation is one completed leader term. GDP growth is defined as the annualized GDP growth rate from when the leader took office until the term ended. Table 4 reports the estimates, and Figure 1 displays the relationship they capture. Promotion is strongly increasing in growth for unconnected leaders and essentially flat for tied leaders.

Hometown tie is an indicator variable that takes the value one if and only if the following scenario happens. During the city leader's term, the central government appoints a new provincial leader to the province in which the city leader works, and the newly appointed provincial leader has the same city of birth as the city leader.[2] Unlike in the U.S., the vast majority of city leaders are not local to the cities they govern. Between 2003 and 2015, only 5% of city leaders were born in the same city they managed. Only 0.5% of leaders had a hometown tie and were local. As mentioned, city-level Chinese politicians' promotion decisions are made by higher-level communist officials in the same province. The appointments of new provincial leaders during a city leader's term are made confidentially by the central party committee, and are hence unexpected by the city leaders in office. To rule out the possibility that a provincial leader favors his hometown comrades by appointing them endogenously, we restrict the definition of a hometown tie to city leaders who had already assumed office when the tie was established (i.e., the provincial leader is appointed at least one year after the city leader had assumed office). Given the salience of the hometown bond among Chinese officials (Section 2), we conjecture that sharing a hometown with the provincial leader shifts a city leader's promotion prospects, and hence the career incentive he faces. About one in seven

[2]The four direct-controlled municipalities (Beijing, Shanghai, Tianjin, and Chongqing) have no provincial superior, as their party secretaries hold provincial rank and are evaluated and promoted directly by the central leadership. For these four cities we define the hometown tie analogously at the central level, equal to one when a newly appointed Politburo member shares the municipality leader's birthplace. Because their promotion tournament operates at the central level, the baseline outcome samples exclude the four municipalities. Pooling them back in, with ties defined analogously against newly appointed Politburo members, leaves every result essentially unchanged (Appendix Table 12).

leader-terms in our sample experiences a tie (57 of 411 in the promotion sample, Table 1). The estimates that follow are not driven by a sparse handful of treated observations.

Coefficient β_1 indicates the correlation between GDP growth and promotion outcome, in the absence of any hometown tie. β_2 captures the effect of hometown tie on the sensitivity of promotion to GDP growth. A positive β_2 indicates that the effect of GDP on promotion is enhanced in the presence of a hometown tie, but a negative estimate for β_2 means a hometown tie diminishes the effect of GDP growth on promotion. The coefficient β_3 on the hometown-tie indicator captures the tie's own effect on promotion, evaluated at zero GDP growth. A positive β_3 indicates that a hometown tie raises a leader's promotion odds, whereas a negative estimate means it lowers them.

All specifications in Table 4 include a comprehensive set of fixed effects to control for unobserved characteristics that might affect a leader's promotion probability beyond the GDP-promotion linkage. City fixed effects take out level differences across cities in the likelihood that their leaders are promoted. With province-year fixed effects, our estimates compare city leaders evaluated by the same provincial committee in the same year. We further include demographic controls such as age and gender, which previous studies have argued are important factors in promotion decisions (Wang, Zhang and Zhou, 2020), and, to address the concern that the hometown tie might pick up leadership qualities specific to certain regions, birth-province fixed effects. Column (2) adds turnover fixed effects, comparing leaders who assumed their posts in the same year. Column (3) controls for the rank of the city leader, given that leaders of different ranks face different promotion odds, and column (4) further controls for whether the leader ever faced a disciplinary action. The coefficients on GDP growth, the hometown tie, and their interaction remain precisely estimated across all specifications.

Higher GDP growth is associated with a greater likelihood of being promoted. In the absence of a hometown tie, a one-standard-deviation increase in annualized GDP growth (about seven percentage points across leader terms) raises a leader's promotion probability by about 13 percentage points (the standardized coefficient of 0.131 in column 4), large relative to the 35% promotion rate of untied leaders, absent a tie.

A hometown tie substantially reduces the importance of GDP performance in someone's evaluation for promotion. As the estimates in column (4) show, having a hometown tie cuts the sensitivity of promotion to GDP by 0.247 per standard deviation, more than offsetting the 0.131 gradient. In fact, the sum $\beta_1 + \beta_2$ is not statistically distinguishable from zero. The null hypothesis that a hometown tie completely obviates the importance of GDP growth to a leader's promotion outcome is not rejected under a Wald test even at the 0.1 level.

Conditional on performance and the fixed-effect comparisons, a hometown tie raises a leader's promotion odds. The estimate on the hometown-tie dummy is positive and statistically significant in every column, at the 5% level in columns (1) through (3) and at the 1% level in the full specification of column (4). The main effect ($\beta_3 = 0.625$ in column 4) is the tie's effect evaluated at zero GDP growth. Because the interaction is negative, the tie raises promotion odds for any growth record below roughly 18% annualized growth. Averaged over the sample growth distribution, the marginal effect of a tie is $+0.13$. Also, the unconditional promotion rate is in fact lower for tied than untied leaders (0.25 versus 0.35, Table 1). The raw gap reflects composition, which masks the tie's effect, while the regression's fixed effects compare leaders within the same province-year and cohort. Conditional on covariates and fixed effects, a hometown tie has a positive effect in boosting a leader's promotion chances.

To sum up, Table 4 shows that absent a tie, promotion is strongly increasing in GDP growth. Developing the local economy is a rewarding path to advancement. A hometown tie then weakens that sensitivity. The growth-promotion link is statistically indistinguishable from zero once a tie forms. Therefore, hometown tie relaxes the GDP-promotion incentive facing a city leader.

Our findings in the first stage add to a rich literature on Chinese political selection. Li and Zhou (2005) document a growth-promotion link for provincial leaders, while Shih, Adolph and Liu (2012) find little robust performance gradient for Central Committee members, and Jia, Kudamatsu and Seim (2015) find that connections and growth act as complements in provincial-leader promotion. We study prefecture leaders inside the within-province tournament, not provincial leaders. Our connection is a tie to the leader's direct evaluator that forms mid-term through the timing of the superior's appointment, not a pre-existing factional alignment that may itself reflect selection. A tie to one's current evaluator substitutes for the growth performance that evaluator would otherwise rely on, which is what Table 4 illustrates, whereas the complementarity documented for factional ties operates at selection into higher office. The two findings are not in conflict.

The decomposition in Table 5 of this gradient demonstrates which growth the tournament rewards. Splitting the promotion gradient into the additive contributions of the primary, secondary, and tertiary sectors, only the secondary-sector contribution carries a positive, statistically significant promotion gradient (significant at the 1% level and as large as the aggregate gradient itself), and the tie severs that contribution specifically, while the primary and tertiary contributions are neither significantly rewarded nor severed. Splitting the secondary contribution one step further, the construction component within it carries a penalty (a per-standard-deviation gradient of -0.36, significant at the 10% level). The reward is associated with the industrial component of secondary output and not with construction activities. We interpret this finer split as suggestive, since construction is a small and volatile share of GDP observed for only 164 leader terms, but it points the same way as the rest of the evidence. The rewarded margin is thus the industrial sector which carries significant emphasis from government above prefectures during China's industrialization. Section 4.4 traces the post-tie recomposition to show that what rises after a tie is the construction activity that the tournament does not reward.

4.2 The Leader's Policy Response: Land, Prices, and Debt

We trace the leader's response on the allocation of land, land prices, local debt, and house-price growth after a hometown tie forms. We regress each outcome on the hometown-tie status of city i in year t,

$$Y_{i,t} = \beta_0 + \beta_1 Hometown\ Tie_{i,t} + \beta_2 X_{i,t} + \epsilon_{i,t} \tag{4.2}$$

where $X_{i,t}$ collects the controls. The key coefficient is β_1 on the hometown tie.

All specifications include city-term and province-by-year fixed effects, and we then add (log) GDP, (log) resident population, and (log) government in-budget fiscal revenue and expenditure cumulatively. The tie coefficient barely moves at any step, which indicates that neither economic fundamentals nor fiscal conditions drive the results. Our preferred specification adds year-within-term fixed effects (labeled "Turnover FE" in the tables), which are indicators for the year of a leader's tenure in which an observation falls, the first year in office, the second, and so on (Ru, 2018). They absorb any systematic profile that outcomes trace over the course of a term. Including them matters here because a hometown tie can only form once a leader is already in office, so treated observations fall later in a leader's term on average, and without absorbing the within-term profile the tie estimate could partly reflect where in the term a leader stands.

Reverse causality would require the Politburo to select provincial appointees on the basis of a single city's land market, and to do so through the accident of a shared birthplace with that city's leader. Provincial appointments are made centrally, for province- and faction-level reasons, and precede the outcomes we study. Section 4.3.2 tests this exogeneity claim directly by permuting superiors' birthplaces. The remaining concern is anticipation. If land and housing markets moved in advance of an expected hometown tie, our estimates would understate the response. The event-study analysis below addresses this by displaying the dynamic timing directly, allowing us to verify that the movements are in the years following tie formation.

4.2.1 Land Reallocates Toward Residential Use

The most direct thing a leader controls is the land the city supplies. A hometown tie severs the growth-promotion link, so city leaders should tilt land back toward residential use once a tie is present.

Consistent with this, residential land supply, measured as the ratio to total land supply, increases with hometown tie, while industrial land supply decreases. The estimates are notably stable across specifications in Table 6. With no controls (column 1 of each panel), a hometown tie raises the residential share by 5.1 percentage points and lowers the industrial share by 5.9

percentage points. The coefficients barely move as the full control set is added (column 3 of each panel). This insensitivity to controls is reassuring, because the estimate does not depend on conditioning on covariates that are themselves plausibly affected by the political-incentive channel, so it is unlikely to be an artifact of "bad controls." In the preferred specification (column 4 of each panel), the effects are larger, the residential share rises by 6 percentage points and the industrial share falls by about 7 percentage points, and significant at the 5% level. Across city-years the residential land share averages about 50% (with a cross-sectional standard deviation of 17 percentage points) and the industrial share about 32%, measured as shares of supplied building area. A hometown tie moves roughly six to seven percentage points between the two uses, on the order of two-fifths of a standard deviation of either share.

The same reallocation reaches commercial land, the third major use, on a one-year delay. The contemporaneous tie leaves the commercial share unmoved, but its one-year lag lowers that share by about four percentage points (a coefficient of -0.043 in the preferred specification, significant at the 1% level, Panel C of Table 6), a sizable movement against a commercial-share mean near 16%. The residential expansion therefore draws on both non-residential uses, industrial land immediately and commercial land a year later.

We find that the hometown tie reallocates land *across* use types, and we do not detect a change in the *total* quantity of land a city makes available. Table 7 reports a regression of (log) total land supply on hometown tie. Across all specifications, the coefficient on tie is close to zero and statistically indistinguishable from it, at -0.050 with no controls beyond the fixed effects, -0.035 adding (log) GDP and (log) population, -0.035 with the full control set, and -0.035 in the preferred specification with year-within-term fixed effects, all with $|t| \leq 0.42$. The total-supply regression is estimated on essentially the same panel as the share regressions and with the same two-way-clustered standard errors, and its point estimates sit close to zero. The total-supply estimate therefore indicates that the response operates mainly through the mix of land uses with no clearly measured expansion of the overall allocation, and we test the scale-based alternatives (tied leaders draw more transfers, credit, or political capacity to release land) directly in Section 4.3.5.

This compositional pattern bears two implications. First, behavior moves on the relative shares of residential, industrial, and commercial land, not the total quantity released. Second, it weighs against scale-based interpretations in which the tie mainly brings a large expansion of land resources, additional political capacity to release land, or a broad loosening of fiscal discipline, because such mechanisms would predict that total supply expands alongside the compositional shift, but the point estimate is near zero. We do not base this conclusion on the total-supply estimate alone, since its interval is wide, but on the direct tests of the resource channels in Section 4.3.5. Contemporaneously, the tie moves none of fixed-asset investment or paved-road area, while annual GDP growth is marginally higher (+1.5 percentage points, significant at 5%, Appendix Table 14), a first hint of the delayed recomposition documented in Section 4.4. Tied leaders re-optimize the *composition* of land use, while the observed scale of investment and infrastructure does not expand. As we show in Section 4.4, output growth then edges up at a two-year horizon even though the resource base does not expand, which is driven by real estate construction which the promotion tournament does not reward.

Table 8 then investigates how land prices respond when the proposed career incentive channel is shut down. Accompanying increased residential land supply and decreased industrial land supply, the unit price of residential land (per building area) falls while the unit price of industrial land rises. In the full-control specification (column 3 of each panel) the residential land price drops by about 23% (significant at 1%) and the industrial price rises by about 12% (significant at 10%). In the preferred specification (column 4 of each panel) the residential decline is about 19% (significant at 5%) and the industrial increase about 15% (significant at 10%). The residential-price decline is the more direct counterpart to the land-supply reallocation, and it is the price result the randomization placebo of Section 4.3.2 validates. The two non-residential prices move the other way, as a falling supply of each implies. The industrial-price increase is marginally significant. The commercial-price increase, reported in Panel C on the one-year-lagged tie that matches the delayed commercial-share response, is about 18% in the preferred specification but imprecisely estimated.

We also find supporting evidence that relaxed career pressure shows up in land-sale revenue, an important component of local fiscal budgets. A body of work on China's land finance holds that the growth incentives facing local leaders are part of what drives their reliance on land sales (Han and Kung, 2015, He, Nelson, Su, Zhang and Zhang, 2025, Liu and Xiong, 2020, Song and Xiong, 2018). The hometown tie offers a way to probe it. Table 9 shows that a tie lowers total land sales revenue by a coefficient of about -0.21 (significant at the 10 percent level) one year after tie formation. The decline is clearest for industrial land, whose revenue falls by about -0.31 (also significant at the 10 percent level) as the supply of industrial land contracts, while the residential and commercial estimates are negative and sizable but imprecise. The event study of Section 4.2.4 tells the same story, with total revenue plummeting at the two-year horizon (-0.46).

4.2.2 Tied Leaders Curtail Borrowing

The same forces that tilt land toward non-residential uses also drives borrowing, because debt-financed infrastructure is among the most direct ways for a leader to produce the growth that promotion rewards. Relaxing the career incentive therefore slows the accumulation of debt. Because the level of debt is a stock with substantial inertia (the within-city-term AR(1) coefficient is 0.60), we use the one-year-lagged tie. Columns (1) through (3) of Table 10 show that a hometown tie reduces the (log) level of local debt, with a coefficient near -0.32 (about a 28% reduction) that is stable as controls are progressively added, statistically significant at the 5% or 10% level. Adding year-within-term fixed effects to the full-control specification (column 5) leaves the result intact (a coefficient of -0.37, about a 31% reduction, significant at 5%), and it survives the placebo test of Section 4.3.2. Local government debt grew at an average of about 36% per year over our sample period. The reduction that follows a tie, about 31% in the preferred specification, is on the order of a full year of borrowing forgone, the cumulative effect of a leader slowing new borrowing rather than retiring debt already incurred.

The choice of lagged versus contemporaneous tie follows the timing each outcome displays in the event studies. Residential land shares and prices move with the contemporaneous tie, and commercial land share responds with a one-year delay, which is why Panel C of Table 6 reports the lagged tie for commercial and the same delayed decline appears in the event study of Section 4.2.4. The debt level, a stock with substantial inertia, likewise loads on the lag, motivating the lagged-tie specification used for debt in Table 10. Finally, the borrowing cut is specific to the local government's own financing-vehicle debt rather than part of a broader credit contraction. Event studies of in-budget revenue and expenditure and of city-wide bank loan and deposit stocks around the tie show no statistically significant declines. What the forgone borrowing would otherwise have financed, most plausibly off-budget, debt-financed investment by financing vehicles, is not separately observed in our data.

4.2.3 House-Price Growth

Finally, we check how hometown ties affect house-price growth. Table 11 shows the estimated effect on a city's house-price growth rate. The specification uses the contemporaneous tie indicator together with two lags of the dependent variable, since house-price growth is strongly autocorrelated. In columns (1) to (3), (log) GDP, (log) resident population, and government fiscal conditions are added gradually to check the robustness of the estimates. Throughout these specifications, the estimate for hometown tie remains similar in magnitude.

Estimates in Table 11 suggest that the hometown-tie channel is associated with slower house-price growth. Column (3) implies a decline of approximately 5.1 percentage points per year, economically meaningful relative to average house-price growth of about 13 percent a year over the sample period. The coefficient attenuates toward zero and loses statistical significance once year-within-term fixed effects are added (column 4). It does not clear the randomization-inference placebo of Section 4.3.2. And its event-study counterpart shows no significant post-tie effect. We therefore treat the house-price result as suggestive corroboration of the land-supply and residential land price results. The cleaner and more directly measured price evidence is the decline in residential land prices in Table 8, which clears the placebo. The result is not an artifact of our splicing the constant-quality price index of Fang, Gu, Xiong and Zhou (2016) (available through 2012) with the NBS city series thereafter. Restricting the sample to the constant-quality index alone (2003–2012) yields a similar estimate.

4.2.4 Dynamics and Pre-Trends

In this section, we examine the dynamics around hometown tie formation. A pre-existing trend would instead point to reverse causality or anticipation. To see this, we estimate an event study around the year a hometown tie forms in a city. We measure event time at the city level, $\tau = t - t_c^*$, where t_c^* is the year of city c's hometown tie establishment. We neutralize the cross-leader level differences by including city-term fixed effects, which absorb leader-specific level shifts. The comparison group is city-term-years in the same province-year without an active tie.

Specifically, we estimate

$$Y_{i,t} = \sum_{\tau \neq -1} \beta_\tau \, \mathbb{1}\{\text{event time} = \tau\}_{i,t} + \alpha_i + \alpha_{p,t} + \alpha_s + \varepsilon_{i,t}$$

where i indexes city-term, t indexes year, the displayed event-time dummies cover $\tau = -3$ to $+2$ (with $\tau = -1$ omitted as the base), α_i is a city-term fixed effect, $\alpha_{p,t}$ is a province-by-year fixed effect, and α_s is a year-within-term fixed effect. The window is bounded on the post-tie side at $\tau = +2$, because post-tie observations must be active-tie years and a tie disappears when either leader leaves office. Under roughly four-year terms the active-tie sample thins quickly beyond two years. Standard errors are two-way clustered at the city and province-year levels, matching the main tables, and in all figures the $\tau = -1$ base is zero by construction.

A subtle issue is whether the post-tie dummies should pool observations regardless of whether the originating tie is still active. In our data, hometown ties are formed by the appointment of a provincial leader during a city leader's term and end whenever either leader leaves office. Among our treated city-terms, 19% of $\tau = +1$ observations and 32% of $\tau = +2$ observations have already lost the tie. We therefore restrict the post-tie observations ($\tau \in \{0, +1, +2\}$) to those for which the tie is active in the year of measurement. Symmetrically, a pre-tie lead ($\tau < 0$) enters only if no tie is active that year, so that a previously-tied observation cannot enter as a clean pre-period. This is why the dynamic coefficients are mechanically larger than the static regression estimates (the residential-share peak of roughly 22 percentage points at $\tau = +1$, versus 6 percentage points in Table 6). The static coefficient averages the tie's effect over its active years, while the event study isolates the peak response among years still active. We therefore use the event study to show evidence on timing and pre-trends, and use the static regression estimates for magnitude. The alternative version of the event study, with post-tie dummies assigned by event time alone, regardless of tie survival, shows the same shape with smaller magnitudes, as expected (residential share $+0.17$ at $\tau = +1$, debt -0.46 at $\tau = +2$), and flat leads throughout.

We organize the event study into five figures. Figure 2 plots land shares for residential, industrial, and commercial land. Figure 3 plots the (log) sold price per building area for the three land types. Figure 4 plots (log) sold sales revenue. Figure 5 plots (log) local-government debt and Figure 6 house-price growth. Price and sales panels use the sold (auction-clearing) variables and the share panels the supplied shares, as in the main regression tables.

In Figure 2, the residential share shows no pre-trend ($\beta_{-3} \approx 0$, $p = 0.99$, and $\beta_{-2} = 8$ percentage points, $p = 0.49$) and rises after the tie, by 22.2 percentage points at $\tau = +1$ ($p = 0.01$) and 19.5 at $\tau = +2$ ($p = 0.04$). The industrial share moves in the opposite direction post-tie (-10.7 percentage points at $\tau = +2$), but neither its post-tie decline ($p = 0.12$) nor its leads (around -9 percentage points) are precisely estimated. The commercial share shows a marginal positive lead at $\tau = -3$ ($+0.098$, $p = 0.07$) and declines after the tie (-0.117 at $\tau = +1$, $p = 0.002$). The clean residential pre-period is the most direct visual support for the parallel-trends assumption underlying the paper's land-reallocation result. In Figure 3, the (log) sold residential price is flat pre-tie and falls afterward, by 0.30 at $\tau = 0$ ($p = 0.07$) and 0.50 at $\tau = +2$ ($p = 0.03$). The industrial price is imprecise. Its point estimate turns positive by the two-year horizon ($+0.19$ at $\tau = +2$), in the same direction as the regression's industrial-price increase, as Table 8 shows. The commercial price is likewise imprecise in the event study and insignificant in the regression (Panel C of Table 8). Figure 4 adds the sold-revenue dynamics, for total land sales and by type. Total revenue is flat through the year of the tie and drops at the two-year horizon (-0.46 at $\tau = +2$). Among these series, residential revenue shows the largest drop (-1.04 at $\tau = +2$), with leads that are flat and a large one (-1.29 at $\tau = -3$) that reflects a run-up into the tie, while industrial and commercial revenue are imprecise throughout. The sold residential price falls (-0.50 at $\tau = +2$) while sold quantity drifts down only noisily. Recorded revenue eases even as the supplied residential share expands.

Figure 5 summarizes results on local government debt. The debt path shows positive but statistically insignificant leads ($\beta_{-3} = 0.12$, $p = 0.60$, and $\beta_{-2} = 0.20$, $p = 0.38$) and falls after the tie, with -0.54 at $\tau = +1$ ($p = 0.08$) and -0.70 at $\tau = +2$ ($p = 0.03$). The pre-period point estimates drift upward within wide confidence intervals, and the post-tie reversal is sharp, mirroring the one-year-lagged estimate in Table 10.

The house-price-growth event study (Figure 6) is suggestive. The post-tie coefficients are negative but individually insignificant (-0.049 at $\tau = +2$, $p = 0.20$). The path's defining feature, though, is a hump peaking at the omitted $\tau = -1$ base, with negative coefficients on both sides including the deep lead (-0.101 at $\tau = -3$, $p = 0.07$).

4.3 Identification Checks and Alternative Interpretations

This section provides four additional robustness checks: sample selection, a randomization-inference placebo that directly tests the exogeneity of tie formation, a test against other social ties, and heterogeneity by the stakes a leader has in the promotion tournament. We then investigate two alternative interpretations of the pattern.

4.3.1 Sample Selection

We begin with two checks on the baseline sample, confirming that neither of the restrictions we imposed in constructing it drives the results. First, nothing hinges on excluding the four direct-controlled municipalities (Beijing, Shanghai, Tianjin, and Chongqing) from the baseline. Their ties are defined at the central level, and Appendix Table 12 re-estimates the preferred specification of each main result pooling them back in, with every coefficient essentially unchanged in magnitude. Second, re-estimating the main outcomes including the 56 dismissed leaders leaves the residential and industrial land shares, the residential land price, and the debt decline unchanged, while the more delicate commercial-share and industrial-price results weaken once these truncated careers are pooled in (Appendix Table 13).

4.3.2 Placebo Test: Permuting Superiors' Birthplaces

The identifying assumption is that, conditional on our fixed effects, the formation of a hometown tie is exogenous with respect to a city's land and housing trajectory. We test this claim using a randomization-inference test that randomizes the superior's birthplace. We hold fixed the real appointment timing and province of every provincial leader and the real birthplaces of every city leader, and randomly permute provincial leaders' birthplaces among themselves within province. Each permutation is run through the identical tie-construction rule, and we re-estimate for each outcome. We repeat this 500 times to build a null distribution for the coefficient on hometown tie under fake assignments. Permuting within province instead of across provinces is essential because a national reshuffle would mechanically collapse the rate of hometown matches and reject by construction. Permuting within province preserves each province's birthplace marginal distributions exactly and randomizes only which appointment, in which year, carries which hometown (the timing variation our design exploits). The permutations are run on the baseline sample under the preferred specification of each outcome. By construction, feeding the true birthplaces through the identical procedure reproduces the headline estimates of the main tables.

Figure 7 reports the result. For the five core outcomes the true-birthplace estimate (red dashed line) lies entirely outside the placebo distribution. None of the 500 within-province permutations produces a coefficient as large in magnitude as the real one for the residential land share (+0.060), the industrial land share (−0.071), the commercial land share (−0.043, on its one-year lag), the residential land price (−0.205), or local government debt (−0.373), giving placebo p-values below 0.002. House-price growth is the exception. Its true coefficient (−0.051) sits inside the placebo distribution ($p = 0.14$), because even random within-province ties generate a mean coefficient of about −0.035, reflecting a within-term mean-reversion in price growth that the permutation does not break. This is an additional reason to interpret the house-price-growth reduced form results as suggestive.

4.3.3 Alternative Social Ties

A complementary concern is that our hometown-tie variable proxies for generic political connectedness. We re-estimate every main outcome using two alternative social ties recorded in our data, a shared party school (educational institutions for training CCP officials) and a shared undergraduate school. These are alternative political connections in the Chinese setting, and indeed the party-school tie is more common in our sample than the hometown tie (194 versus 103 nonzero city-years), yet, as Appendix Table 17 reports, neither reproduces our pattern of results. The coefficients are small and statistically insignificant. Pooling all three ties into an "any-tie" indicator likewise produces no effect. Only the hometown tie removes the sensitivity of promotion to GDP performance. The tie×GDP-growth interaction in the promotion regression is -0.25 per standard deviation ($t = -2.6$) for hometown ties but small and statistically insignificant for both the party-school ($+0.07$, $t = 0.8$) and original-school (-0.05, $t = -0.5$) ties (reported in the note to Table 17). Because only the hometown tie relaxes the GDP-promotion incentive, only it propagates to land allocation, prices, and debt. This pattern is difficult to reconcile with any account in which our results reflect generic political connectedness, or a social tie that merely happens to coincide with the timing of appointments.

4.3.4 Heterogeneity: The Effect Tracks Promotion Stakes

If hometown tie works through career incentives, then tie's effects are likely concentrated among leaders who are genuinely competing for promotion, and muted for those effectively out of the tournament. We test this by interacting the hometown tie with an indicator for leaders who have passed the mandatory age ceiling for their rank (a start-of-term age above 55 for prefecture-rank officials, 60 for sub-provincial, and 65 for provincial, for whom further promotion is largely foreclosed). Appendix Table 18 is consistent with the prediction. The effect on every outcome is at least as large for high-stakes leaders as in the pooled sample, and the point estimates attenuate for out-of-tournament leaders. That the strongest response is governed by whether a leader can still be promoted supports the interpretation that the operative force is the GDP-promotion career incentive.

4.3.5 Alternative Interpretations

The first alternative interpretation concerns the active price management documented by Chang, Wang and Xiong (2026), who show that during the post-2020 downturn local governments propped up residential land and house prices through countercyclical land purchases by their financing vehicles and limits on new-home sales permits, producing a divergence between prices and transaction volumes. A potential concern is that our lower residential prices and lower debt are simply a tied leader easing off such support. Their instruments manage the price level of residential land and housing. Our central result is the composition of primary land supply, with the residential share rising and the industrial share falling while the total quantity released does not move (Table 7). Supporting prices, whether by buying land or by rationing sales permits, shifts land demand, not the industrial-residential mix. The industrial land price rises as the residential price falls. And the real-economy recomposition of Section 4.4, with foreign investment and industrial employment down, housing sold to households up, and output holding up, lies outside of a pure price-management account. Where the two accounts meet is in treating land and land-backed debt as strategic levers. Chang, Wang and Xiong (2026) identify instruments through which local governments manage prices, while we identify the political-career incentive that governs how a leader allocates land and incurs debt in the first place.

The second alternative is preferential treatment from above. A newly arrived superior who favors his hometown comrade might channel resources toward the tied city, through fiscal transfers, favorable project and credit allocation, or laxer oversight of borrowing, and the downstream outcomes could then reflect treatment from above. Because cities in our period could not borrow within the general budget, general-budget expenditure in excess of own revenue, the revenue a city raises itself, is financed by transfers from above, and because every such transfer reaches the city through the province, the province's own funds together with the central funds it allocates, the ratio of expenditure to own revenue captures the margin a provincial leader controls. Land-sale proceeds do not enter this ratio, because they accrue to the separate governmental-fund budget. The margin is wide, with expenditure running roughly 1.7 times own revenue over our period, yet the tie has an estimated zero effect on this ratio at every horizon from five years before the tie to two years after (Appendix Figure 10). Project money corroborates this at the funding-source level. We measure it with static regressions not event studies, because the Urban Construction Statistical Yearbook begins only in 2006 and leaves the year-by-year leads on these series too thinly populated to be reliable. No source of municipal-infrastructure investment rises in tied city-years (Appendix Table 15), with total infrastructure funding and central appropriations carrying negative point estimates and the local-appropriation line that bundles provincial project money with the city's own funds flat. Total bank loans, loan growth, and credit intensity are likewise unchanged, as are the total land supply the city makes available (Table 7) and fixed investment and paved roads (Appendix Table 14). Laxer oversight of borrowing predicts more debt, but the debt stock falls by about 31 percent, and resource channeling predicts more industrial activity, whereas foreign investment and industrial employment growth decline (Section 4.4). Two further patterns are difficult to reconcile with a treatment-from-above story. The effects attenuate once the leader ages past the promotion-eligibility ceiling (Section 4.3.4), although a provincial leader's resources have no obvious reason to follow the client's eligibility clock, and other social ties that plausibly carry generic favoritism, such as a shared alma mater, move none of the outcomes (Section 4.3.3).

4.4 From Policy Levers to the Real Economy: The Recomposition of Growth

This section shows that hometown ties influence local economic development. Because real activity responds to land allocation only after land is built upon and put to use, we measure these outcomes with the same city-level event study and within-province placebo developed above, plotting effects at the two-year horizon post tie. The real-economy series are drawn from the CEIC prefecture panel described in Section 3. Figure 8 plots the dynamics on the same $\tau \in [-3, +2]$ window as the land-market figures. Tests of the displayed leads fail to reject flat pre-trends for any outcome (p-values from 0.19 to 0.95).

First, the city's industrial growth slows down. Foreign direct investment falls on every measure we observe. Utilized FDI drops by 0.35 log points at $\tau = +2$, contracted FDI by 0.58, and the number of FDI contracts by 0.16, clearing the within-province placebo at the 5% level for the first two measures and the 10% level for the third (placebo $p = 0.040$, 0.002, and 0.054). The industrial workforce contracts. Growth in secondary-sector employment falls (-0.099, placebo $p = 0.01$) and the city's non-agricultural population, its stock of urban, largely industrial workers, declines (-0.058, $p = 0.03$).

Second, residential floor space sold per capita rises by 0.16 log points by $\tau = +2$. This is the welfare-relevant counterpart of the residential land reallocation, because housing is typically the dominant asset on household balance sheets in emerging economies, China included (Badarinza, Campbell and Ramadorai, 2016, Badarinza, Balasubramaniam and Ramadorai, 2019), so the supply of housing to households is where the land reallocation would matter for household welfare. It clears the placebo marginally ($p = 0.09$), and the broader set of household-welfare outcomes we examined, including disposable income, consumption, and fertility, does not move.

Third, aggregate output does not fall and even rises. The rise is a recomposition, because construction and the output it induces drive it, not the industrial effort that promotion values. GDP growth increases by 0.05 (placebo $p = 0.004$) by the second post-tie year. The city produces at least as much, but less foreign-investment-driven, labor-intensive industry, and more residential and construction activity. The natural reconciliation of a rising secondary sector with falling secondary-sector employment is that construction, which sits inside secondary industry, expands as residential land is built out while labor-intensive manufacturing contracts. Using the provincial-yearbook decomposition of city GDP by industry (Section 3), on the common set of cities for which the secondary-sector split is observed (87), construction value added grows fastest after a tie (a coefficient of about 0.20), while the non-construction remainder of secondary value added rises only mildly (about 0.07) and tertiary value added is flat (Figure 9). Also, investment and infrastructure do not rise on impact (Appendix Table 14) and we detect no increase in total land supply (Table 7). Finally, we find no support for spillovers to untied cities. The relative comparison does not reflect discouraged competitors, because untied cities in the same province show no change in growth, foreign investment, or land composition when a province-mate becomes tied (Appendix Table 16).

5 Conclusion

In this paper, we provide reduced-form causal evidence that the career incentives created by China's GDP-based promotion system are a first-order determinant of how local officials allocate land and how much debt their cities take on. Exploiting the plausibly exogenous timing of provincial and central leadership appointments, and the hometown ties they create, we show that when career incentives are relaxed, city leaders reallocate land toward residential use, lower residential land prices, and curtail local government debt. We do not detect a change in the total quantity of land supplied, and direct tests of transfers, credit, and project allocation from above come back null. Suggestive evidence also shows house-price growth slows in tied cities. A randomization-inference placebo that permutes superiors' birthplaces confirms that the land-reallocation, land-price, and debt results are not artifacts of the appointment process. In the real economy, we find that the reallocation recomposes local growth. The industrial growth that the tournament rewards recedes, residential housing sold expands, and aggregate output does not fall and may even rise. Career incentives shape the composition of local growth, steering officials toward the activity a promotion tournament emphasizes.

Our findings bear clear policy implications. The criteria by which local officials are evaluated and promoted matter for China's land, housing, and local-debt outcomes. Measures to contain the housing boom and the buildup of local government debt that focus only on credit conditions or demand-side instruments may miss a fundamental driver. We echo recent policy analysis arguing that derisking China's real estate sector requires replacing the financing model that ties local governments to land revenue and debt (Xiong, 2025, Chang, Wang and Xiong, 2026).

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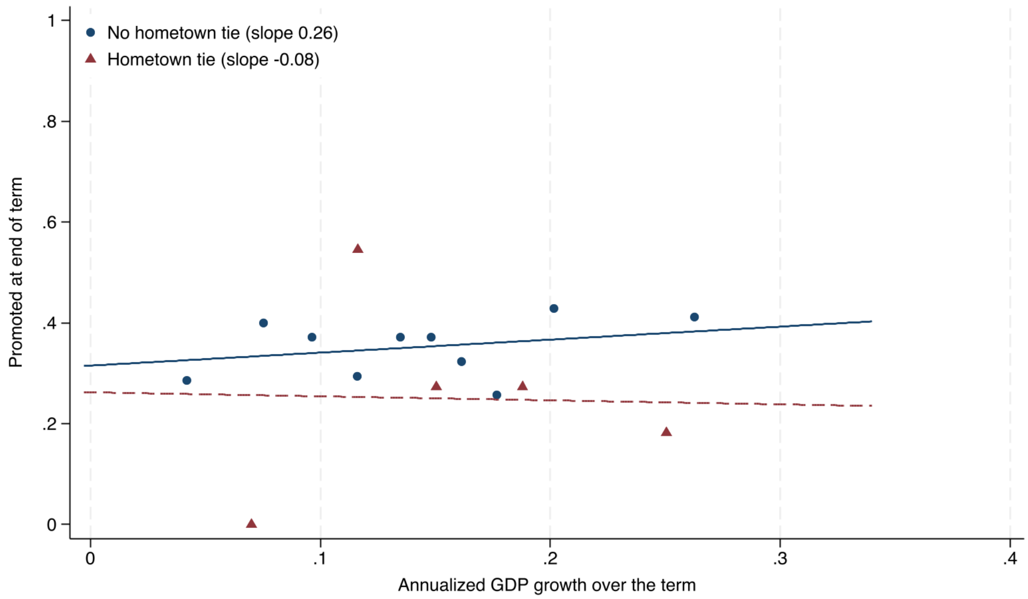
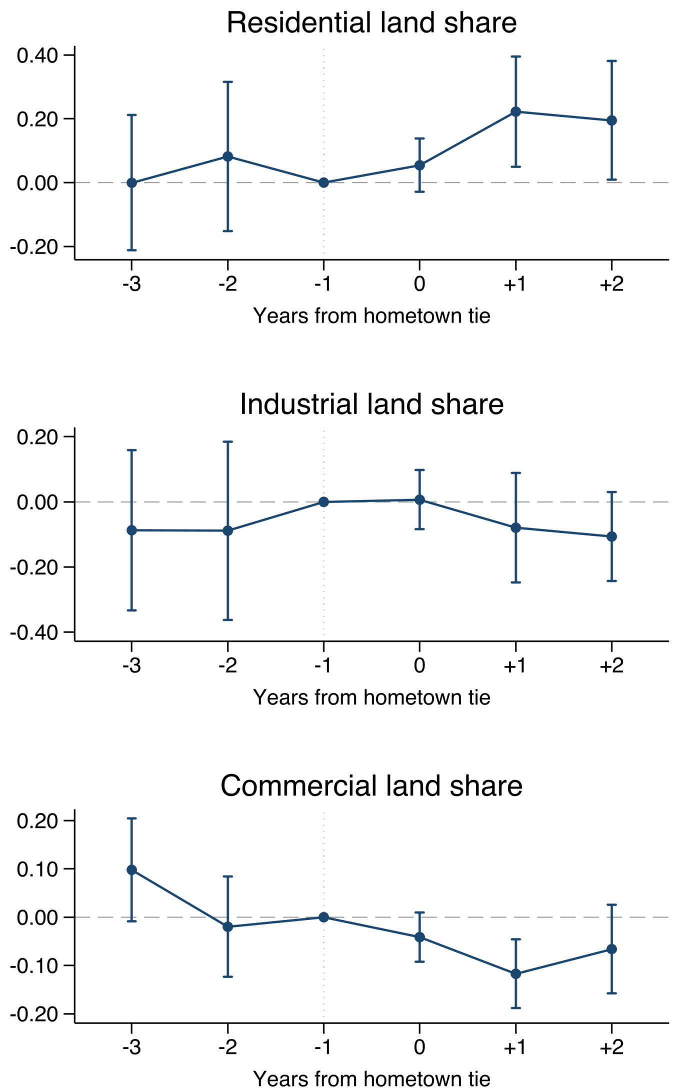
6 Figures

Figure 1: Promotion, GDP Growth, and the Hometown Tie



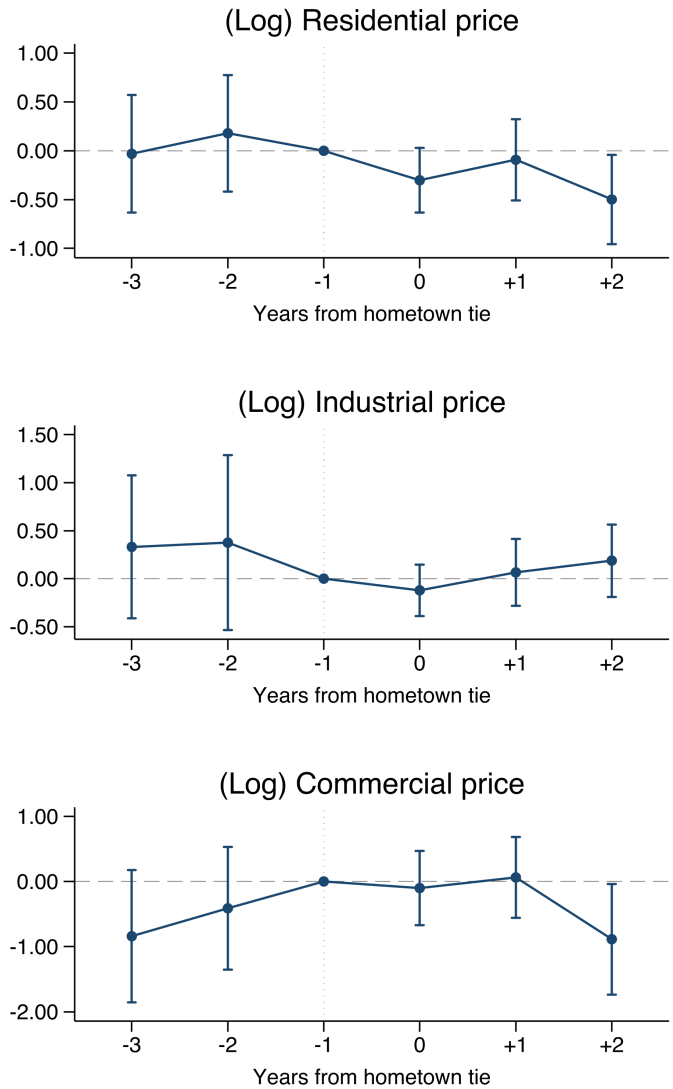
Notes. Binned means of the end-of-term promotion indicator against annualized GDP growth over the term, separately for leader terms without (circles, ten bins) and with (triangles, five bins) a hometown tie, on the promotion estimation sample of Table 4 (growth trimmed at the 1st and 99th percentiles for display). Lines are unconditional linear fits estimated on the underlying terms.

Figure 2: Event Study: Planned Land Shares by Type



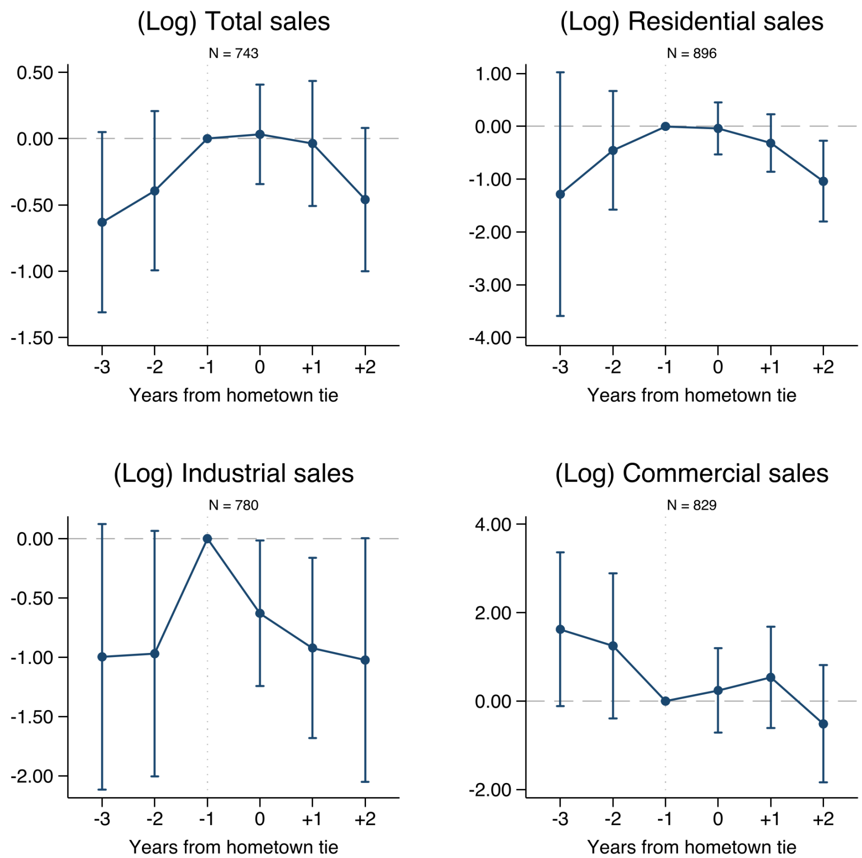
Notes. Event-study coefficients β_τ from separate regressions of the supplied land share on event-time dummies, τ years from a city's first hometown tie, for residential, industrial, and commercial land. The omitted base is $\tau = -1$. Specifications include city-term, province-by-year, and year-within-term fixed effects, with post-tie years restricted to active ties, leads to no-tie years. Standard errors are two-way clustered at the city and province-year levels. Bars are 95% confidence intervals. Source: CREIS.

Figure 3: Event Study: (Log) Sold Land Price by Type



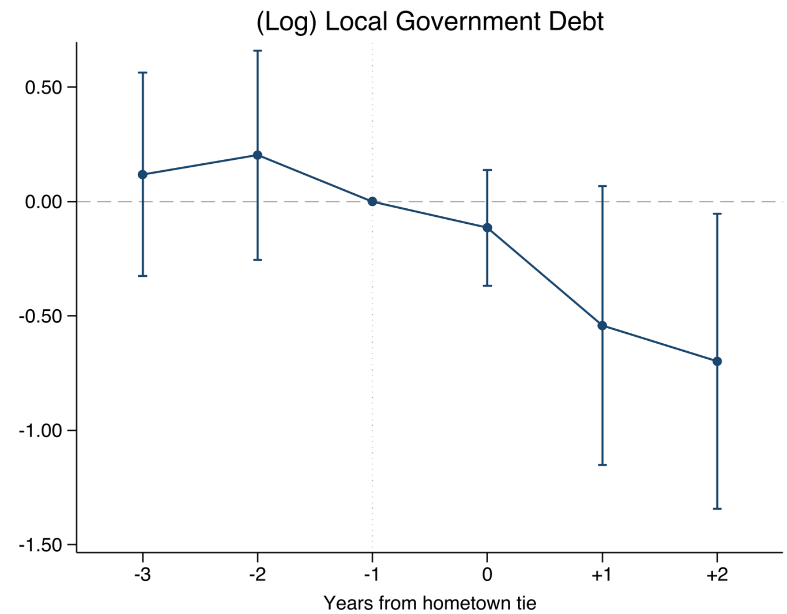
Notes. Event-study coefficients for the log sold price per building area, by land type. Specifications, fixed effects, controls, and standard-error clustering match Figure 2, except that the commercial-price panel reports city-clustered standard errors. The omitted base is $\tau = -1$. Source: CREIS.

Figure 4: Event Study: (Log) Land Sales Revenue, Total and by Type



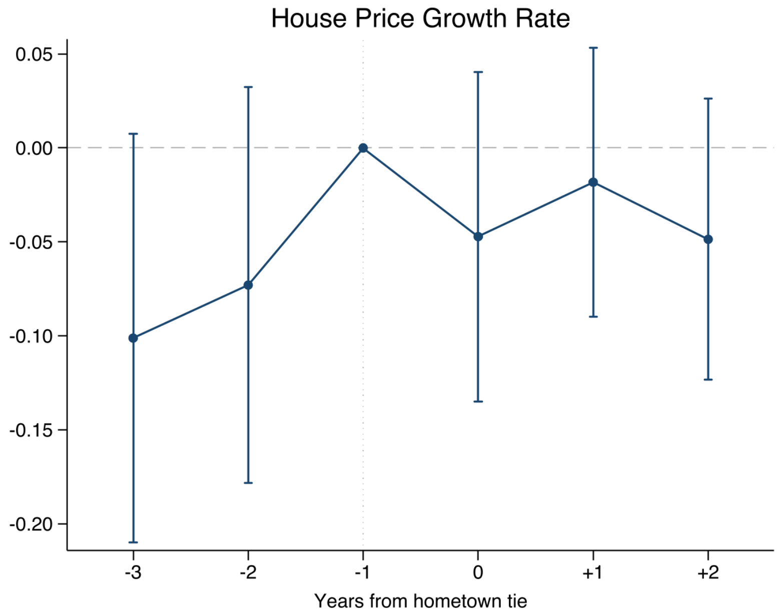
Notes. Event-study coefficients for the log sold sales revenue (price $\times$ quantity), for total land sales (top-left panel) and by land type. Specifications, fixed effects, and standard-error clustering match Figure 2. The omitted base is $\tau = -1$. Source: CREIS.

Figure 5: Event Study: Local Government Debt



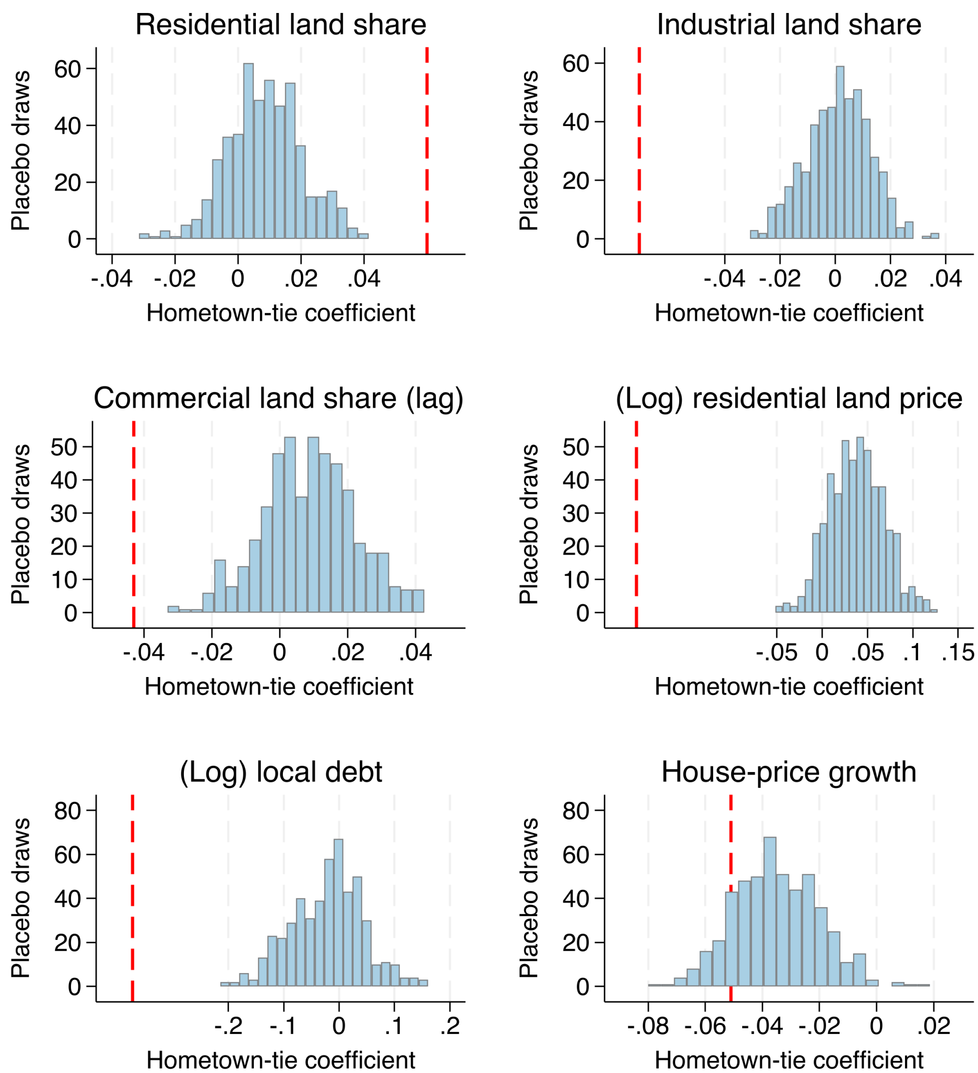
Notes. Event-study coefficients around a city's first hometown tie for (log) local government debt. The omitted base is $\tau = -1$. Specifications match Figure 2, including city-term, province-by-year, and year-within-term fixed effects, with post-tie years restricted to active ties and leads to no-tie years. Standard errors are two-way clustered at the city and province-year levels, with 95% confidence intervals shown. Local debt is the local-government financing-vehicle debt panel of Huang, Pagano and Panizza (2020) (Section 3).

Figure 6: Event Study: House-Price Growth



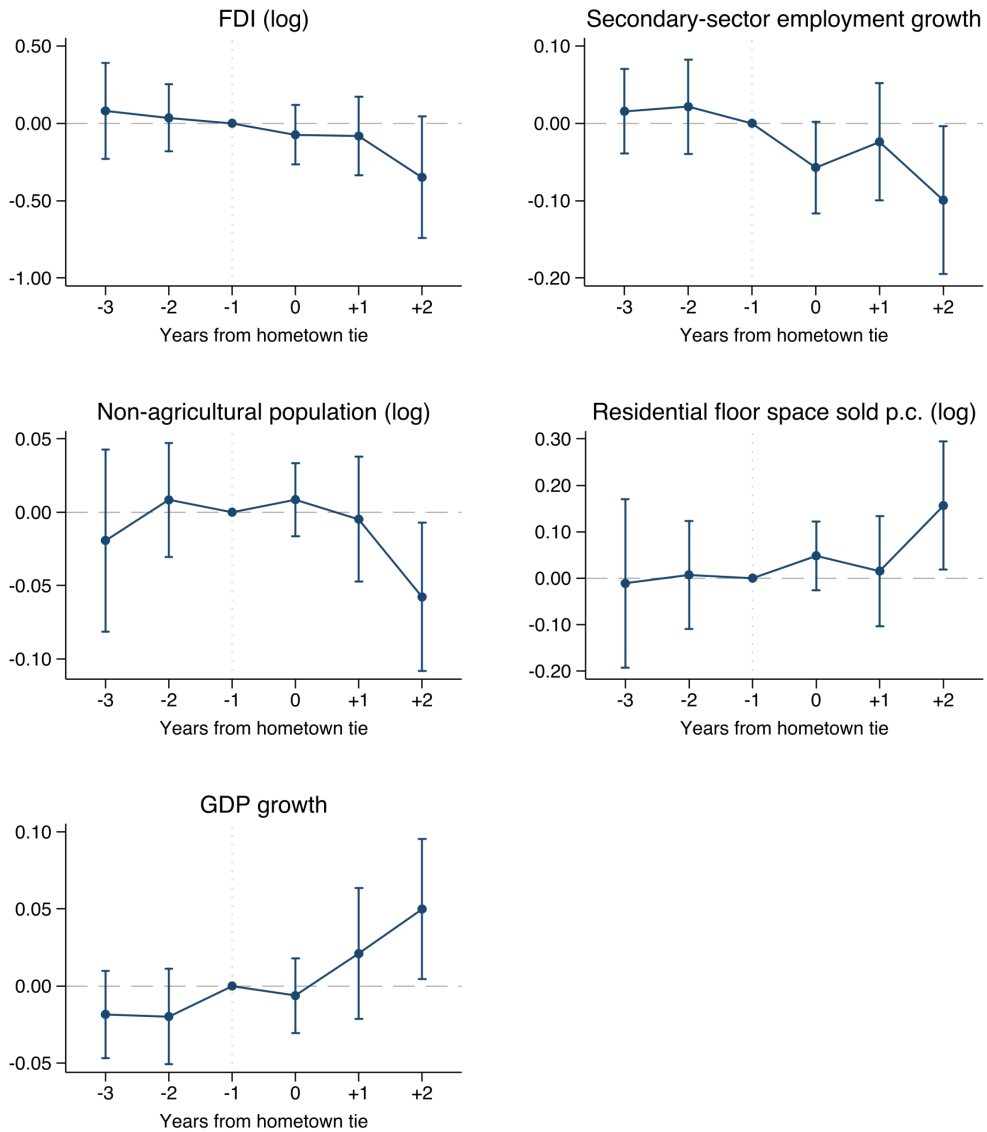
Notes. Event-study coefficients around a city's first hometown tie for house-price growth. The omitted base is $\tau = -1$. Specifications match Figure 2, including city-term, province-by-year, and year-within-term fixed effects, with post-tie years restricted to active ties and leads to no-tie years. Standard errors are two-way clustered at the city and province-year levels, with 95% confidence intervals shown. House-price growth is the annual log change in the spliced constant-quality price index (Section 3).

Figure 7: Placebo Test: Permuting Provincial Leaders' Birthplaces Within Province



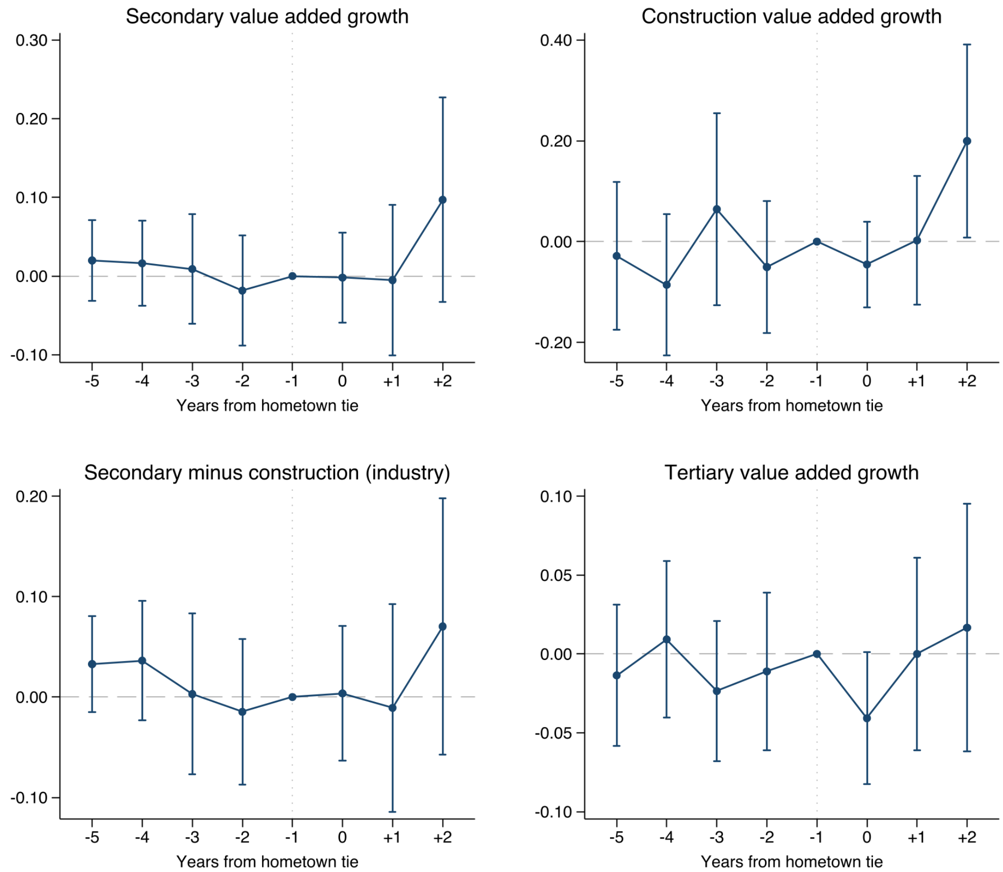
Notes. Each panel plots the distribution of the estimated hometown-tie coefficient across 500 placebo samples in which provincial leaders' birthplaces are randomly permuted within province. Each draw uses the same tie-construction rule and preferred specification as the corresponding outcome, including city-term, province-by-year, and year-within-term fixed effects for the land and debt outcomes, and the lagged-dependent-variable specification without year-within-term fixed effects for house-price growth. Standard errors are two-way clustered throughout. The red dashed line marks the estimate under leaders' true birthplaces, equal to the corresponding headline estimate. Two-sided placebo p-values are below 0.002 for the residential, industrial, and commercial land shares, residential land price, and local debt, and 0.14 for house-price growth. The sample is the baseline sample of the main tables.

Figure 8: The Real Economy Recomposes Around the Hometown Tie



Notes. Event-study coefficients around a city's first hometown tie for five real-economy outcomes, on the same city-term, province-by-year, and year-within-term fixed effects and binned specification as Figure 2, with post-tie years restricted to active ties and leads to no-tie years. The displayed window is $\tau \in [-3, +2]$ with $\tau = -1$ omitted as the base. Each outcome's two-year ($\tau = +2$) coefficient is tested against a within-province randomization-inference placebo that permutes superiors' birthplaces across 500 draws, with p-values reported in the text. Top row: utilized FDI (log, the panel titled "FDI") and secondary-sector employment growth. Middle row: non-agricultural population (log) and residential floor space sold per capita (log). Bottom: GDP growth. Standard errors two-way clustered at the city and province-year levels, with 95% confidence intervals shown. Source: CEIC prefecture statistics merged to the analysis sample.

Figure 9: Decomposing the Post-Tie Output Rise by GDP Component



Notes. Event-study coefficients for annual value-added growth by GDP component, on the same city-term, province-by-year, and year-within-term fixed effects and binned specification as Figure 8, with $\tau = -1$ omitted as the base. The four panels are secondary value-added growth, construction value-added growth, the non-construction remainder of secondary value added, and tertiary value-added growth, from the provincial-yearbook decomposition of city GDP (Section 3). All four are estimated on the common set of 87 cities for which the construction split is observed. The panels are comparable. Bars are model-based 95% confidence intervals with standard errors two-way clustered at the city and province-year levels. Source: EPS provincial yearbooks merged to the analysis sample.

7 Tables

Table 1: Summary Statistics of City Party Secretaries, by Term

	No Hometown Tie			Hometown Tie		
	Mean	Std.	count	Mean	Std.	count
Promotion	0.35	0.48	354	0.25	0.43	57
Term Length	4.01	1.59	354	4.79	1.58	57
Term Start Age	49.97	3.56	354	49.58	4.11	57

Notes. Summary statistics for the 411 completed leader terms in the promotion estimation sample of Table 4 (city party secretaries in office 2003–2015), reported separately for terms with and without a hometown tie. Promotion equals one when the city party secretary moves to a higher rank after his city term. Term length is years in the term, and term start age is age at term start. Source: China Political Elite Dataset.

Table 2: Summary Statistics of the Land-Market Variables

	Unit	N	Mean	Std. dev.	P25	Median	P75
Residential share of planned supply	share	630	0.50	0.17	0.39	0.51	0.61
Industrial share of planned supply	share	630	0.32	0.16	0.21	0.31	0.43
Commercial share of planned supply	share	630	0.16	0.10	0.09	0.14	0.21
Total planned supply (building area)	10k sq. m	630	1948	1618	808	1547	2576
Sold residential land price	RMB/sq. m	1057	1331	2826	490	757	1330
Sold industrial land price	RMB/sq. m	890	303	322	180	238	319

Notes. Summary statistics on the baseline estimation panel of city-year observations of 179 prefecture cities, 2003–2015, excluding the four direct-controlled municipalities and the dismissed leaders (Section 4). Shares are supplied building-area shares by use type. Total land supply is in 10,000 square meters of building area, and prices are sold (auction) prices per square meter, CPI-adjusted. Sample sizes vary with outcome availability. Source: China Real Estate Index System.

Table 3: Summary Statistics of Macro Variables, by City

	Unit	N	Mean	Std. dev.	Min	Median	Max
Housing price	RMB/sq m	195	3,750.41	2,166.56	1,657.70	3,017.30	16,364.43
Fixed investment	RMB mn	195	105,752.95	103,832.76	9,653.58	66,654.79	679,984.85
GDP	RMB tn	195	0.1826	0.2245	0.0146	0.1047	1.7508
ΔGDP	RMB tn	195	0.0168	0.0196	0.0008	0.0097	0.1200
Average wage	RMB	195	31,342.38	7,339.60	16,316.36	29,810.69	70,256.63
Paved road	sq m bn	195	0.02	0.03	0.00	0.01	0.26
Household registration	Person th	195	4,769.80	3,278.12	553.39	3,915.89	32,745.60
Usual residence	Person th	186	5,023.02	3,519.75	633.81	4,332.87	28,922.84
Govt. revenue	RMB bn	195	15.82	31.97	0.77	7.03	313.28
Govt. expenditure	RMB mn	195	23,486.48	36,146.12	3,989.12	14,843.67	349,434.69
Deposits	RMB mn	195	298,465.74	655,356.56	23,897.36	108,967.86	6,200,904.56

Notes. Summary statistics for city-level economic data for 195 cities, 2004–2015, except usual-residence population, available for 186 of them. Source: China National Bureau of Statistics.

Table 4: Promotion, GDP, and Hometown Tie

	Dependent variable: end-of-term promotion			
	(1)	(2)	(3)	(4)
GDP Growth (s.d.)*Hometown Tie	-0.246**	-0.275***	-0.244**	-0.247**
	(0.101)	(0.105)	(0.098)	(0.096)
Hometown Tie	0.529**	0.604**	0.574**	0.625***
	(0.229)	(0.247)	(0.235)	(0.227)
Annualized GDP Growth (s.d.)	0.124***	0.160***	0.146***	0.131***
	(0.046)	(0.048)	(0.046)	(0.043)
Prov-Year FE	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Turnover FE	N	Y	Y	Y
Gender FE	Y	Y	Y	Y
Age FE	Y	Y	Y	Y
Rank FE	N	N	Y	Y
Termination FE	N	N	N	Y
N	411	411	411	411
Adj. R^2	0.0991	0.0381	0.0705	0.0951

Notes. Linear probability model of end-of-term promotion over 411 completed city party-secretary terms, 2003–2015 (Section 3). All columns absorb birth-province fixed effects. Annualized GDP growth is standardized to unit standard deviation. Its coefficient and interaction with the tie are per-standard-deviation effects, while the hometown-tie main effect is in natural units. Standard errors, two-way clustered at the city and province-year levels, in parentheses.

Table 5: Decomposing the Promotion Gradient by GDP Component

GDP component contribution	Reward gradient (per s.d.)	Gradient × Tie (per s.d.)	Hometown Tie (main effect)
Aggregate GDP	0.14***	−0.16	0.49**
	(0.05)	(0.10)	(0.25)
Primary contribution	0.03	−0.07	0.18
	(0.05)	(0.10)	(0.17)
Secondary contribution	0.13***	−0.29***	0.59***
	(0.05)	(0.10)	(0.18)
of which: construction	−0.36*	−0.11	0.08
	(0.19)	(0.28)	(0.23)
Tertiary contribution	0.09	−0.06	0.22
	(0.07)	(0.14)	(0.34)

Notes. Dependent variable is end-of-term promotion over completed leader terms, 2003–2015, excluding the four direct-controlled municipalities. Each sector's contribution to aggregate annualized GDP growth sums to aggregate growth by construction. The reward-gradient and interaction columns standardize each contribution to unit standard deviation, so they are per-standard-deviation effects, comparable across rows, while the final column reports the hometown-tie main effect in natural units. Specification, fixed effects, and two-way clustering match column (4) of Table 4, and the aggregate row is the EPS counterpart of that GDP gradient. $*$, $**$, $***$ denote significance at the 10%, 5%, and 1% levels.

Table 6: Hometown Tie and Land Supply Ratio

	Panel A. Residential land share				*Panel B. Industrial land share*				*Panel C. Commercial land share*			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hometown Tie	0.0513*	0.0472*	0.0464*	0.0600**	-0.0594*	-0.0530*	-0.0493*	-0.0710**	-0.0372**	-0.0373***	-0.0374***	-0.0428***
	(0.0290)	(0.0257)	(0.0258)	(0.0258)	(0.0320)	(0.0276)	(0.0287)	(0.0319)	(0.0152)	(0.0142)	(0.0141)	(0.0155)
(Log) GDP		0.378	0.316	0.366		-0.335	-0.210	-0.214		-0.170	-0.279	-0.330
		(0.329)	(0.346)	(0.359)		(0.266)	(0.269)	(0.285)		(0.260)	(0.327)	(0.313)
(Log) Population		-0.218	-0.250	-0.296		-0.367	-0.342	-0.324		0.509**	0.502**	0.539**
		(0.200)	(0.203)	(0.184)		(0.249)	(0.245)	(0.252)		(0.227)	(0.209)	(0.229)
(Log) Fiscal Revenue			0.0227	0.0476			-0.182	-0.175			0.225*	0.193*
			(0.137)	(0.130)			(0.139)	(0.144)			(0.115)	(0.112)
(Log) Fiscal Expenditure			0.107	0.108			-0.0314	-0.0461			-0.0732	-0.0449
			(0.122)	(0.125)			(0.124)	(0.130)			(0.0617)	(0.0641)
City-Term FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Turnover FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
N	485	485	485	485	485	485	485	485	456	456	456	456
Adj. R^2	0.472	0.471	0.468	0.468	0.502	0.508	0.509	0.501	0.256	0.285	0.292	0.304

Notes. OLS estimates of the hometown-tie effect on the share of land supply by use type. Panel A is the residential share, Panel B the industrial share, and Panel C the commercial share. Columns (1)–(4) use the same specifications within each panel, adding controls progressively, with column (4) the preferred specification. The tie row reports contemporaneous Hometown Tie in Panels A and B and one-year-lagged Hometown Tie in Panel C, since commercial land responds with a one-year delay. Sample is up to 191 prefecture-level cities, 2003–2015 (excluding the four direct-controlled municipalities). Standard errors, two-way clustered at the city and province-year levels, are in parentheses.

Table 7: Hometown Tie and Total Land Supply

	Dependent variable: (log) total land supply			
	(1)	(2)	(3)	(4)
Hometown Tie	-0.0501	-0.0350	-0.0345	-0.0347
	(0.119)	(0.124)	(0.128)	(0.129)
(Log) GDP		-0.882	-1.078	-1.134
		(0.808)	(0.813)	(0.762)
(Log) Population		-1.172	-1.350	-1.190
		(1.001)	(0.946)	(0.937)
(Log) Fiscal Revenue			-0.152	-0.225
			(0.428)	(0.423)
(Log) Fiscal Expenditure			0.627	0.740
			(0.485)	(0.513)
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	N	N	N	Y
N	516	516	516	516
Adj. R^2	0.830	0.831	0.831	0.829

Notes. OLS estimates of the hometown-tie effect on the (log) level of total land supply. Sample is up to 191 prefecture-level cities, 2003–2015 (excluding the four direct-controlled municipalities). Each observation is a city-year pair. Standard errors, two-way clustered at the city and province-year levels, in parentheses.

Table 8: Hometown Tie and Unit Price of Land by Type

	Panel A. (Log) residential price				*Panel B. (Log) industrial price*				*Panel C. (Log) commercial price*			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Hometown Tie	-0.258***	-0.263***	-0.266***	-0.205**	0.127*	0.119*	0.113*	0.141*	0.100	0.0912	0.0910	0.168
	(0.0955)	(0.0907)	(0.0930)	(0.104)	(0.0716)	(0.0637)	(0.0644)	(0.0713)	(0.102)	(0.0921)	(0.0945)	(0.108)
(Log) GDP		0.131	0.0400	-0.127		0.803	0.524	0.458		1.401	0.835	0.840
		(0.790)	(0.764)	(0.744)		(0.581)	(0.562)	(0.560)		(1.198)	(1.272)	(1.173)
(Log) Population		0.707	0.718	0.719		-0.0740	-0.154	-0.104		-0.0729	-0.123	-0.0871
		(1.255)	(1.278)	(1.220)		(0.660)	(0.676)	(0.722)		(0.449)	(0.446)	(0.363)
(Log) Fiscal Revenue			0.177	0.286			0.257	0.236			0.837	1.076
			(0.456)	(0.457)			(0.328)	(0.339)			(0.728)	(0.718)
(Log) Fiscal Expenditure			-0.0727	-0.251			0.194	0.175			-0.0456	-0.385
			(0.523)	(0.581)			(0.267)	(0.269)			(0.627)	(0.608)
City-Term FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Turnover FE	N	N	N	Y	N	N	N	Y	N	N	N	Y
N	506	506	506	506	506	506	506	506	468	468	468	468
Adj. R^2	0.860	0.860	0.858	0.858	0.688	0.686	0.685	0.678	0.685	0.683	0.683	0.689

Notes. OLS estimates of the hometown-tie effect on the (log) sold price of land per unit building area. Panel A is residential land, Panel B industrial land, and Panel C commercial land. Columns (1)–(4) use the same specifications within each panel, adding controls progressively, with column (4) the preferred specification. The tie row reports contemporaneous Hometown Tie in Panels A and B and one-year-lagged Hometown Tie in Panel C, matching the delayed commercial-share response. Sample is up to 191 prefecture-level cities, 2003–2015 (excluding the four direct-controlled municipalities). In the no-control column of each panel, city-clustered standard errors are reported (Cameron, Gelbach and Miller, 2011). Standard errors, two-way clustered at the city and province-year levels, are in parentheses.

Table 9: Hometown Tie and Land Sales Revenue

	(1) Total	(2) Residential	(3) Industrial	(4) Commercial
L.Hometown Tie	-0.207*	-0.257	-0.305*	-0.155
	(0.119)	(0.175)	(0.165)	(0.209)
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	Y	Y	Y	Y
N	676	772	699	733
R^2	0.952	0.924	0.876	0.880

Notes. Estimates of (log) land sales revenue on the one-year-lagged hometown tie (L.Hometown Tie), matching the lagged specification used for the local-debt stock and the commercial land share. Column (1) is total land sales revenue, and columns (2)–(4) decompose it by use type. All specifications include the full control set with city-term, province-year, and year-within-term fixed effects. Sample is 2003–2015, excluding the four direct-controlled municipalities and dismissed leaders. Each observation is a city-year pair. Source: China Real Estate Index System. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Hometown Tie and Local Government Debt

	Dependent variable: (log) local government debt			
	(1)	(2)	(3)	(4)
L.Hometown Tie	-0.325*	-0.330**	-0.323*	-0.373**
	(0.164)	(0.164)	(0.167)	(0.162)
Average Bond Coupon	0.0574**	0.0606**	0.0561*	0.0480
	(0.0288)	(0.0292)	(0.0296)	(0.0310)
(Log) GDP		-1.259	-1.482*	-1.661*
		(0.843)	(0.887)	(0.891)
(Log) Population		-0.853***	-0.977**	-0.989**
		(0.323)	(0.386)	(0.460)
(Log) Fiscal Revenue			0.134	-0.120
			(0.390)	(0.333)
(Log) Fiscal Expenditure			0.339	0.402
			(0.289)	(0.263)
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	N	N	N	Y
N	359	359	359	359
Adj. R^2	0.965	0.965	0.965	0.965

Notes. OLS estimates of the hometown-tie effect on (log) local government outstanding debt. Columns (1)–(3) use the one-year-lagged tie with controls added progressively. Column (4) is a distributed-lag specification with the contemporaneous, one-year-lagged, and two-year-lagged tie. Column (5) adds year-within-term fixed effects to the full-control specification. Sample is up to 191 prefecture-level cities, 2003–2015 (excluding the four direct-controlled municipalities). Each observation is a city-year pair. Standard errors, two-way clustered at the city and province-year levels, in parentheses.

Table 11: Hometown Tie and House Price Growth Rate

	Dependent variable: house-price growth			
	(1)	(2)	(3)	(4)
Hometown Tie	-0.0559**	-0.0493**	-0.0514**	-0.0304
	(0.0213)	(0.0215)	(0.0226)	(0.0232)
L.House-price Growth	-0.394***	-0.408***	-0.397***	-0.401***
	(0.102)	(0.0991)	(0.0994)	(0.0954)
L2.House-price Growth	-0.235***	-0.250***	-0.248***	-0.257***
	(0.0610)	(0.0620)	(0.0631)	(0.0642)
(Log) GDP		-0.394**	-0.350	-0.362
		(0.193)	(0.221)	(0.224)
(Log) Population		0.443	0.432	0.396
		(0.574)	(0.537)	(0.562)
(Log) Fiscal Revenue			0.0793	0.114
			(0.111)	(0.109)
(Log) Fiscal Expenditure			-0.169	-0.152
			(0.119)	(0.120)
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	N	N	N	Y
N	386	386	386	386
Adj. R^2	0.421	0.428	0.425	0.409

Notes. OLS estimates of the hometown-tie effect on house-price growth. Sample is up to 191 prefecture-level cities, 2003–2015 (excluding the four direct-controlled municipalities). Each observation is a city-year pair. Standard errors, two-way clustered at the city and province-year levels, in parentheses.

Appendices

A Robustness: Pooling the Direct-Controlled Municipalities and Including Dismissed Leaders

Table 12: Main Results Pooling the Four Direct-Controlled Municipalities

Panel A. Land shares and debt				
	Resid. Share	Indust. Share	Comm. Share	(Log) Debt
Hometown Tie	0.0706***	-0.0769***		
	(0.0261)	(0.0292)		
L.Hometown Tie			-0.0431***	-0.376**
			(0.0146)	(0.155)
Controls	Y	Y	Y	Y
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	Y	Y	Y	Y
N	503	503	474	378
Panel B. Land prices and house-price growth				
	Resid. Price	Indust. Price	Comm. Price	HP Growth
Hometown Tie	-0.178*	0.140**		-0.0437**
	(0.101)	(0.0671)		(0.0217)
L.Hometown Tie			0.199*	
			(0.108)	
Controls	Y	Y	Y	Y
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	Y	Y	Y	N
N	522	522	484	413

Notes. The main land, price, and debt results re-estimated pooling back in the four direct-controlled municipalities (Beijing, Shanghai, Tianjin, and Chongqing), with their ties defined analogously at the central level. Panel A reports the supplied residential, industrial, and commercial land shares and (log) local government debt, and Panel B the (log) sold residential, industrial, and commercial land prices and house-price growth. The commercial-share, commercial-price, and debt columns use the one-year-lagged tie, as in their main tables, while the rest use the contemporaneous tie. Each carries the full control set and the fixed effects of its main table, indicated in the lower rows of each panel. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Main Results Including the 56 Dismissed Leaders

Panel A. Land shares and debt				
	Resid. Share	Indust. Share	Comm. Share	(Log) Debt
Hometown Tie	0.0668**	-0.0776**		
	(0.0256)	(0.0311)		
L.Hometown Tie			-0.0239	-0.299**
			(0.0195)	(0.150)
Controls	Y	Y	Y	Y
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	Y	Y	Y	Y
N	561	561	543	435
Panel B. Land prices and house-price growth				
	Resid. Price	Indust. Price	Comm. Price	HP Growth
Hometown Tie	-0.208**	0.0459		-0.0273
	(0.0927)	(0.0796)		(0.0221)
L.Hometown Tie			0.0889	
			(0.115)	
Controls	Y	Y	Y	Y
City-Term FE	Y	Y	Y	Y
Prov-Year FE	Y	Y	Y	Y
Turnover FE	Y	Y	Y	N
N	584	584	560	572

Notes. The main land, price, and debt results re-estimated including the 56 city leaders dismissed during their careers, whom the baseline analysis excludes. Panel A reports the supplied residential, industrial, and commercial land shares and (log) local government debt, and Panel B the (log) sold residential, industrial, and commercial land prices and house-price growth. The commercial-share, commercial-price, and debt columns use the one-year-lagged tie, as in their main tables, while the rest use the contemporaneous tie. Each carries the full control set and the fixed effects of its main table, indicated in the lower rows of each panel. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B Aggregate Outcomes

Table 14: Hometown Tie and Aggregate Real-Economy Outcomes

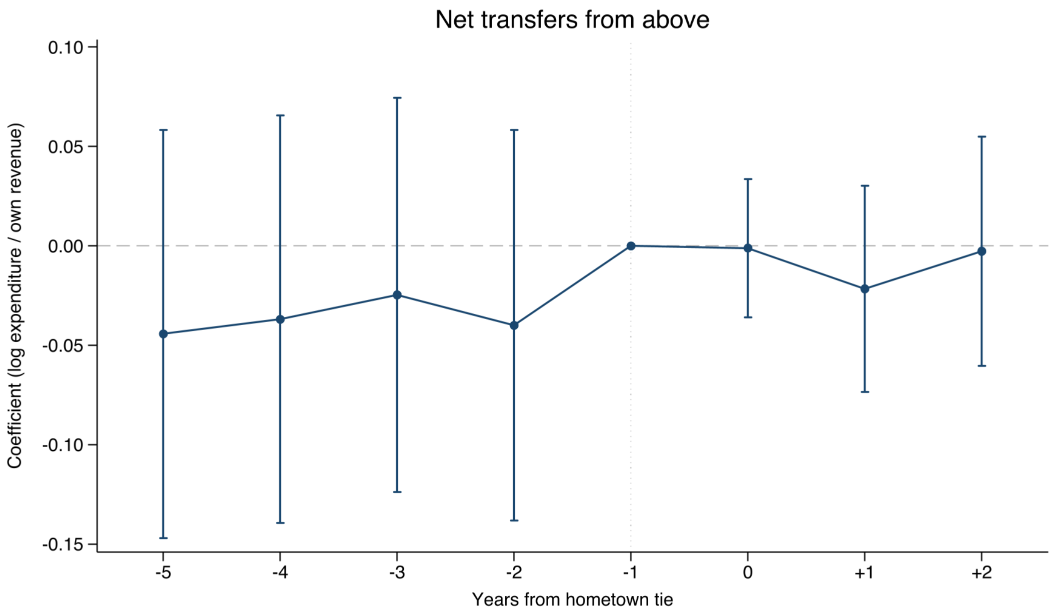
	(1)	(2)	(3)
	Δlog GDP	(Log) Fixed Inv.	(Log) Paved Road
Hometown Tie	0.0154**	0.0254	-0.00548
	(0.00640)	(0.0240)	(0.0216)
City-Term FE	Y	Y	Y
Prov-Year FE	Y	Y	Y
Turnover FE	Y	Y	Y
N	1365	1592	1587
Adj. R^2	0.558	0.992	0.975

Notes. Each column regresses an aggregate real-economy outcome on the contemporaneous hometown tie, controlling for (log) resident population, (log) government in-budget revenue and expenditure, and (log) GDP, with city-term, province-year, and year-within-term fixed effects. The GDP control is omitted from column (1), whose outcome is GDP growth. The outcomes are annual GDP growth (Δlog GDP), (log) fixed-asset investment, and (log) paved-road area. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C Preferential Treatment From Above

The output result of Section 4.4 could in principle reflect a provincial patron channeling resources to the tied city. The two exhibits do not support it.

Figure 10: Resources From Above Do Not Respond to the Hometown Tie



Notes. Event-study coefficients around a city's first hometown tie for net transfers from above, estimated on the full analysis panel. The plotted outcome is the log of the general-budget expenditure-to-own-revenue ratio, which proxies transfers from above. The window runs from $\tau = -5$ to $\tau = +2$. The event clock, binned specification, fixed effects, and two-way clustering otherwise match Figure 8, with $\tau = -1$ the omitted base and 95% confidence intervals shown. Source: analysis panel, general-budget expenditure and revenue.

Table 15: Resources From Above Do Not Respond: Static Tests

Funding channel	Hometown tie	s.e.	N
Net transfers (log expenditure / own revenue)	-0.012	(0.012)	1,592
Total infrastructure funding / expenditure (log)	-0.214^{*}	(0.123)	1,097
Central appropriation received (any, 0/1)	-0.195^{*}	(0.101)	1,097
Central share of infrastructure funding	-0.041	(0.026)	1,097
Local + provincial appropriation / expenditure (log)	-0.133	(0.150)	1,031

Notes. Each row is a separate static regression of the indicated funding channel on the contemporaneous hometown tie, with (log) GDP and (log) population controls and province-by-year, city-term, and year-within-term fixed effects. Funding series are from the Urban Construction Statistical Yearbook (2006–2015) and cover 169 to 183 cities. The comprehensive net-transfer test is plotted in Figure 10. Two-way clustered (city, province-year) standard errors in parentheses. $^{*}\, p < 0.1$.

Table 16: Untied Province-Mates Do Not Respond When a Province-Mate Becomes Tied

	(1) GDP growth	(2) FDI (log)	(3) Resid. share
Province-mate tied	−0.006	0.006	0.018
	(0.005)	(0.056)	(0.014)
one-year lag	−0.001	0.040	0.018
	(0.007)	(0.068)	(0.015)
two-year lag	0.001	−0.001	0.023
	(0.006)	(0.069)	(0.016)
Joint p-value	0.522	0.946	0.306
City FE	Y	Y	Y
Year FE	Y	Y	Y
N	1,178	1,165	517
R^2	0.460	0.916	0.637

Notes. The sample is untied city-years, and the regressor is an indicator for any province-mate being tied that year, entered with its first two annual lags. The dependent variables are annual GDP growth, log utilized foreign investment, and the residential share of land supply. Specifications absorb city and year fixed effects, and standard errors are clustered by province.

D Specificity: Hometown Ties versus Other Social Ties

Table 17: The Effect Is Specific to Hometown Ties

	Resid. share	Indust. share	Resid. land price	(Log) debt	HP growth
Hometown tie	0.060**	-0.071**	-0.205**	-0.373**	-0.051**
	(0.025)	(0.032)	(0.104)	(0.162)	(0.023)
Party-school tie	-0.031	0.002	0.064	0.094*	0.034*
	(0.023)	(0.025)	(0.063)	(0.056)	(0.020)
Original-school tie	-0.020	-0.003	0.022	-0.133*	-0.034
	(0.020)	(0.031)	(0.113)	(0.076)	(0.049)

Notes. Each cell reports the coefficient on a hometown, shared-party-school, or shared-original-school tie from the preferred specification for each outcome (full controls, city-term, province-year, and year-within-term fixed effects). Outcome samples are harmonized as in the corresponding main tables. The debt column uses the one-year-lagged tie, and the house-price-growth column omits within-term fixed effects and includes two lags of the dependent variable. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

E Heterogeneity by Promotion Stakes

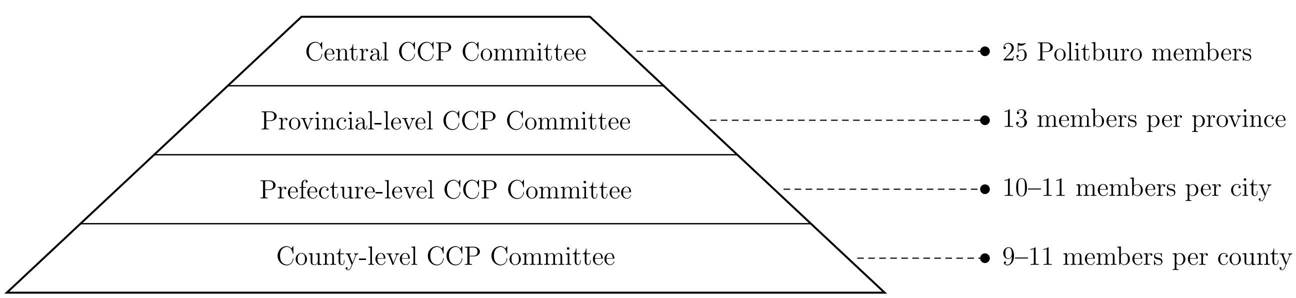
Table 18: The Effect Is Concentrated Among Leaders Still in the Promotion Tournament

	Resid. share	Indust. share	Resid. land price	(Log) debt
Hometown tie	0.061**	-0.078**	-0.210**	-0.474**
	(0.029)	(0.036)	(0.105)	(0.206)
× out-of-tournament	-0.011	0.050	0.142	0.377*
	(0.060)	(0.036)	(0.190)	(0.196)
Implied: out-of-tournament	0.051	-0.027	-0.068	-0.097*
	(0.050)	(0.022)	(0.170)	(0.052)

Notes. Each column adds to the preferred specification an interaction between the hometown tie and an indicator for leaders out of the promotion tournament, who are past the customary age ceiling for their rank (start-of-term age above 55 for prefecture, 60 for sub-provincial, and 65 for provincial rank). The first row is the tie effect for leaders still in the tournament, the second the differential for out-of-tournament leaders, and the third their implied effect (the sum). The debt column uses the one-year-lagged tie. Controls, fixed effects (city-term, province-year, year-within-term), and two-way clustering match the preferred columns of the main tables. Standard errors, two-way clustered at the city and province-year levels, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

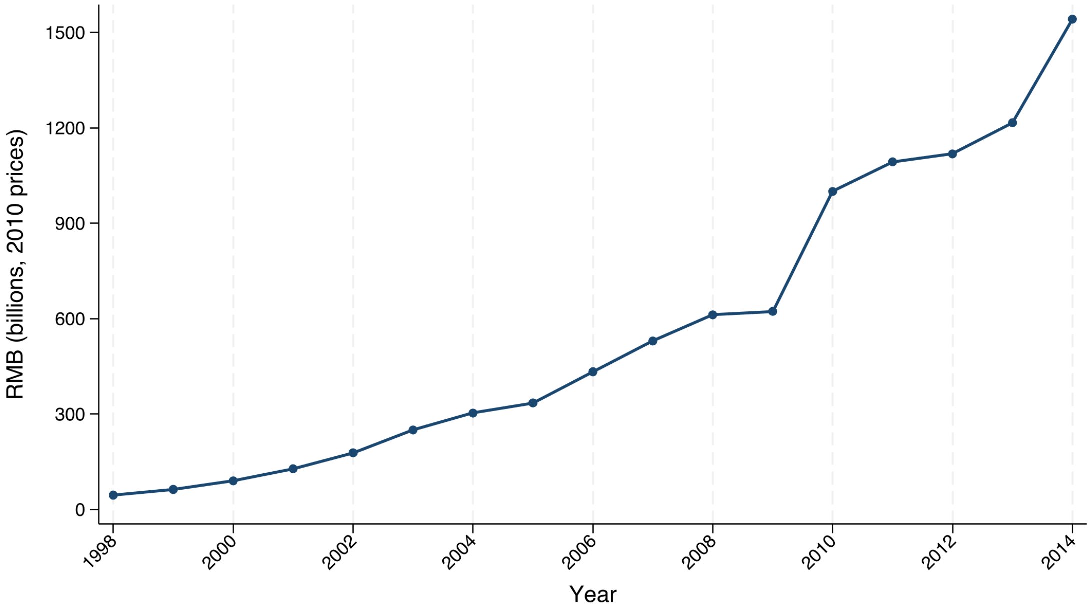
F Additional Descriptive and Institutional Figures

Figure 11: The Political Pyramid of the Communist Party of China



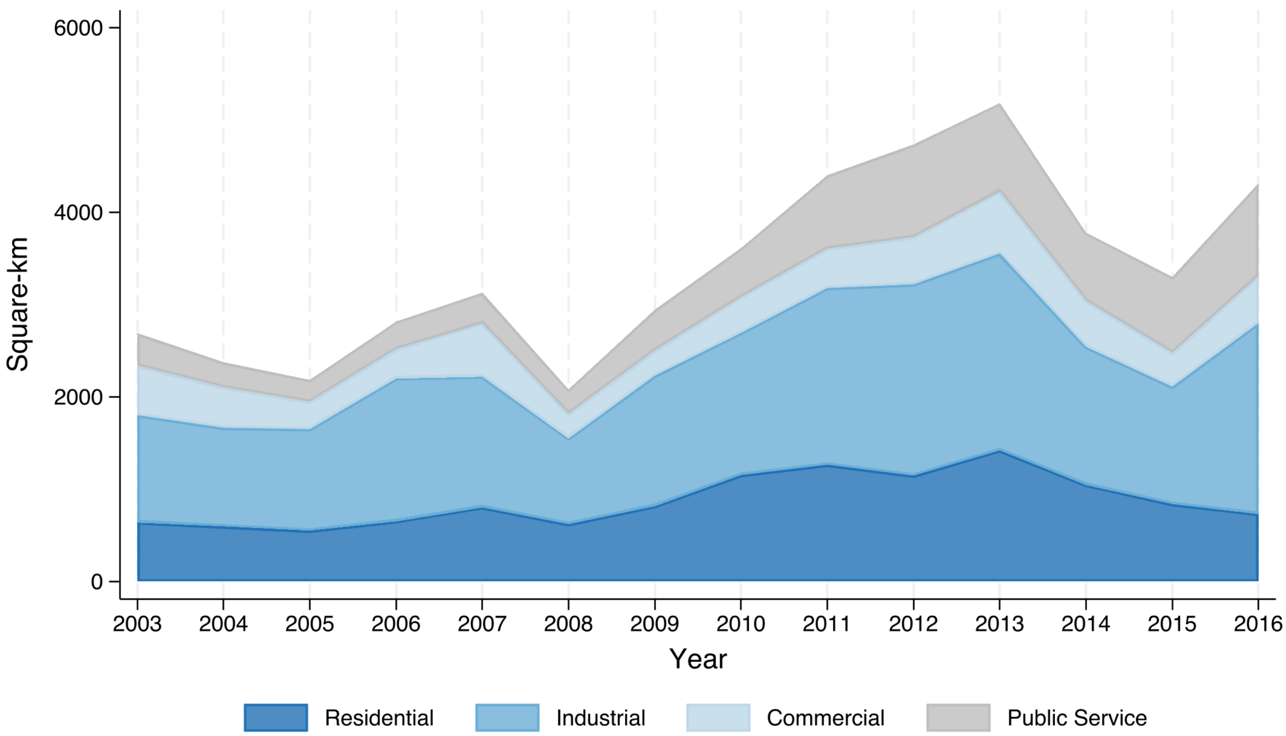
Notes. This graph plots the organizational structure of the Communist Party of China.

Figure 12: Total Land Sales to Real Estate Developers (RMB Billions)



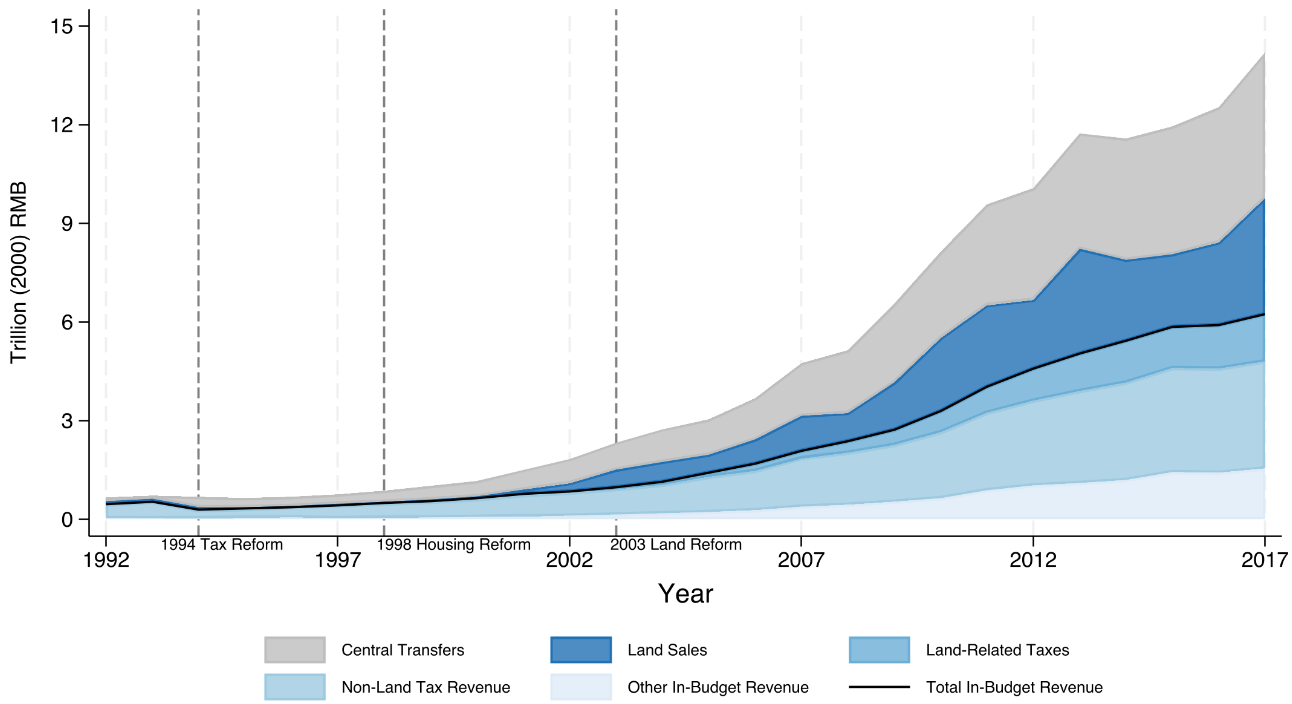
Notes. Annual total land sales revenue to real estate developers in the Chinese land market, in RMB billions, CPI-adjusted (2010=1). Source: China Statistics Yearbook (1999–2016).

Figure 13: Land Allocation by Use Type



Notes. Composition of land supplied in China, 2003–2016. Omitted categories are land for water facilities, transportation, and special purpose. Source: China National Bureau of Statistics.

Figure 14: Local Governments' Financial Structure



Notes. Local Chinese governments' financial structure, 1992–2017. Source: land sales data from Zhang (2009) for 1992–1999 and Finance Yearbook of China for 2000–2017, government budget data from China National Bureau of Statistics.